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Uncertainty Quantification for Inverse Modelling

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1 Introduction

2 The Fisher Information approach

In the context of statistical estimation for dynamic systems, Fisher Information Matrix (FIM)-based methods provide a powerful framework for quantifying uncertainty in parameter estimates. By leveraging the sensitivity of model outputs to its parameters, FIM allows us to construct confidence intervals that indicate the likely range of true parameter values and prediction intervals that capture the expected variability in future observations. These intervals are essential for assessing model reliability and guiding experimental design, particularly in complex or nonlinear systems where intuition alone may be misleading. The details of constructing FIM-based confidence and prediction intervals are provided in the following subsections.

2.1 FIM-based confidence intervals

The construction of confidence intervals based on the Fisher Information Matrix (FIM) provides a principled way to quantify uncertainty in parameter estimates obtained from maximum likelihood estimation. Under regularity conditions, the inverse of the FIM approximates the covariance matrix of the estimators, allowing confidence intervals to be derived from their asymptotic normality. This framework offers a computationally efficient and theoretically grounded approach for assessing estimation precision, especially in complex or high-dimensional models.

More formally, let $\ell(\boldsymbol{\theta})$ denote the log-likelihood of a statistical model, with parameter vector

$$\boldsymbol{\theta} = (\theta_1, \dots, \theta_p)' \in \Theta \subseteq \mathbb{R}^p. \quad (1)$$

Let $\hat{\boldsymbol{\theta}}$ be the maximum-likelihood estimate (MLE). Then, the following holds.

Proposition 1. *The quadratic Fisher-information approximation for the log-likelihood profile of a single component θ_j of $\boldsymbol{\theta} \in \Theta \subseteq \mathbb{R}$ is given by*

$$\ell_p(\theta_j) \approx \ell(\hat{\boldsymbol{\theta}}) - \frac{1}{2} I_{jj|-j} (\theta_j - \hat{\theta}_j)^2, \quad (2)$$

where $I_{jj|-j}$ is the conditional curvature of the profile likelihood, that is

$$I_{jj|-j} = I_{jj} - I_{j,-j} I_{-j,-j}^{-1} I_{-j,j} = \frac{1}{(I^{-1})_{jj}}. \quad (3)$$

which equals

$$I_{jj|-j} = \frac{1}{(I^{-1})_{jj}}. \quad (4)$$

Proof. Let $\ell(\boldsymbol{\theta})$ be the log-likelihood and $\hat{\boldsymbol{\theta}}$ the maximum likelihood estimate (MLE). We consider a second-order Taylor expansion of $\ell(\boldsymbol{\theta})$ around $\hat{\boldsymbol{\theta}}$:

$$\ell(\boldsymbol{\theta}) \approx \ell(\hat{\boldsymbol{\theta}}) + (\boldsymbol{\theta} - \hat{\boldsymbol{\theta}})' \frac{\partial \ell}{\partial \boldsymbol{\theta}} + \frac{1}{2} (\boldsymbol{\theta} - \hat{\boldsymbol{\theta}})' \frac{\partial^2 \ell}{\partial \boldsymbol{\theta} \partial \boldsymbol{\theta}'} (\boldsymbol{\theta} - \hat{\boldsymbol{\theta}}). \quad (5)$$

Since the gradient vanishes at the MLE, we have

$$\ell(\boldsymbol{\theta}) \approx \ell(\hat{\boldsymbol{\theta}}) - \frac{1}{2} (\boldsymbol{\theta} - \hat{\boldsymbol{\theta}})' \mathbf{I} (\boldsymbol{\theta} - \hat{\boldsymbol{\theta}}), \quad (6)$$

where $\mathbf{I} = -\ell''(\hat{\boldsymbol{\theta}})$ is the observed Fisher information. Partition $\boldsymbol{\theta} = (\theta_j, \boldsymbol{\theta}_{-j})$ and correspondingly partition the information matrix as

$$\mathbf{I} = \begin{pmatrix} I_{jj} & I_{j,-j} \\ I_{-j,j} & I_{-j,-j} \end{pmatrix}. \quad (7)$$

The quadratic form then becomes

$$(\boldsymbol{\theta} - \hat{\boldsymbol{\theta}})' \mathbf{I} (\boldsymbol{\theta} - \hat{\boldsymbol{\theta}}) = I_{jj}(\theta_j - \hat{\theta}_j)^2 + 2(\theta_j - \hat{\theta}_j)I_{j,-j}(\boldsymbol{\theta}_{-j} - \hat{\boldsymbol{\theta}}_{-j}) + (\boldsymbol{\theta}_{-j} - \hat{\boldsymbol{\theta}}_{-j})' I_{-j,-j}(\boldsymbol{\theta}_{-j} - \hat{\boldsymbol{\theta}}_{-j}). \quad (8)$$

The profile log-likelihood for θ_j is obtained by maximizing over $\boldsymbol{\theta}_{-j}$:

$$\ell_p(\theta_j) = \max_{\boldsymbol{\theta}_{-j}} \ell(\theta_j, \boldsymbol{\theta}_{-j}). \quad (9)$$

Setting the derivative with respect to $\boldsymbol{\theta}_{-j}$ to zero yields

$$I_{-j,-j}(\boldsymbol{\theta}_{-j} - \hat{\boldsymbol{\theta}}_{-j}) + I_{-j,j}(\theta_j - \hat{\theta}_j) = 0 \implies \boldsymbol{\theta}_{-j}^* = \hat{\boldsymbol{\theta}}_{-j} - I_{-j,-j}^{-1} I_{-j,j}(\theta_j - \hat{\theta}_j). \quad (10)$$

Substituting $\boldsymbol{\theta}_{-j}^*$ back into the quadratic form gives

$$\begin{aligned} (\boldsymbol{\theta} - \hat{\boldsymbol{\theta}})' \mathbf{I} (\boldsymbol{\theta} - \hat{\boldsymbol{\theta}}) &= I_{jj}(\theta_j - \hat{\theta}_j)^2 - 2(\theta_j - \hat{\theta}_j)^2 I_{j,-j} I_{-j,-j}^{-1} I_{-j,j} + (\theta_j - \hat{\theta}_j)^2 I_{j,-j} I_{-j,-j}^{-1} I_{-j,j} \\ &= (\theta_j - \hat{\theta}_j)^2 (I_{jj} - I_{j,-j} I_{-j,-j}^{-1} I_{-j,j}). \end{aligned} \quad (11)$$

Define the *conditional Fisher information* (curvature) as

$$I_{jj|-j} = I_{jj} - I_{j,-j} I_{-j,-j}^{-1} I_{-j,j}. \quad (12)$$

Hence, the quadratic approximation to the profile log-likelihood is

$$\ell_p(\theta_j) \approx \ell(\hat{\boldsymbol{\theta}}) - \frac{1}{2} I_{jj|-j} (\theta_j - \hat{\theta}_j)^2, \quad (13)$$

as claimed. Lastly, let $\mathbf{I}$ be a symmetric positive definite matrix, partitioned as

$$\mathbf{I} = \begin{pmatrix} I_{jj} & I_{j,-j} \\ I_{-j,j} & I_{-j,-j} \end{pmatrix}, \quad (14)$$

where I_{jj} is a scalar, $I_{j,-j}$ is a $1 \times (p-1)$ row vector, $I_{-j,j} = I_{j,-j}'$, and $I_{-j,-j}$ is a $(p-1) \times (p-1)$ matrix. The Schur complement of $I_{-j,-j}$ in $\mathbf{I}$ is defined as

$$I_{jj|-j} := I_{jj} - I_{j,-j} I_{-j,-j}^{-1} I_{-j,j}. \quad (15)$$

A standard result in block matrix algebra states that if

$$\mathbf{I}^{-1} = \begin{pmatrix} A & B \\ C & D \end{pmatrix}, \quad (16)$$

then

$$A = (\text{Schur complement of } I_{-j,-j})^{-1} = (I_{jj|-j})^{-1}. \quad (17)$$

In other words,

$$(I^{-1})_{jj} = \frac{1}{I_{jj|-j}} \implies I_{jj|-j} = \frac{1}{(I^{-1})_{jj}}. \quad (18)$$

Therefore, we have

$$\boxed{I_{jj|-j} = I_{jj} - I_{j,-j} I_{-j,-j}^{-1} I_{-j,j} = \frac{1}{(I^{-1})_{jj}}}. \quad (19)$$

□

2.2 FIM-Based Prediction Intervals

Fisher Information Matrix (FIM)-based prediction intervals provide a principled approach to quantify how **parameter uncertainty propagates into uncertainty about future predictions**. Since the FIM captures the local curvature of the log-likelihood, its inverse approximates the **asymptotic covariance** of the maximum likelihood estimator (MLE). This allows us to express prediction uncertainty as a quadratic function of the design or input variable, yielding interpretable intervals that widen in regions of low information.

More formally, let $\theta \in \mathbb{R}^p$ be a parameter vector, and let the log-likelihood based on observations $Y_1, \dots, Y_n$ be

$$\ell(\theta; Y_1, \dots, Y_n). \quad (20)$$

The maximum likelihood estimator (MLE) is defined by

$$\hat{\theta} = \arg \max_{\theta} \ell(\theta). \quad (21)$$

Let

$$\mathcal{J}(\theta) = -\mathbb{E} \left[\frac{\partial^2 \ell(\theta)}{\partial \theta \partial \theta'} \right] \quad (22)$$

be the Fisher Information Matrix. Under standard regularity conditions, the MLE satisfies asymptotic normality, that is

$$\hat{\theta} \stackrel{\text{approx}}{\sim} N(\theta, \mathcal{J}(\theta)^{-1}). \quad (23)$$

Consider a scalar function

$$\eta = g(\theta), \quad g : \mathbb{R}^p \rightarrow \mathbb{R}. \quad (24)$$

of the parameters θ . A first-order Taylor expansion of $g(\hat{\theta})$ around θ gives

$$g(\hat{\theta}) \approx g(\theta) + \nabla_{\theta} g(\theta)' (\hat{\theta} - \theta), \quad (25)$$

where $\nabla_{\theta} g(\theta)$ denotes the gradient of g with respect to θ . Then, by applying the delta method, the approximate variance of $g(\hat{\theta})$ is

$$\text{Var}(g(\hat{\theta})) \approx \nabla_{\theta} g(\hat{\theta})' \mathcal{J}(\hat{\theta})^{-1} \nabla_{\theta} g(\hat{\theta}). \quad (26)$$

Let Y_* denote a future observation with conditional distribution

$$Y_* | \theta \sim p(y_* | \theta), \quad (27)$$

and suppose its conditional mean is

$$\mathbb{E}[Y_* | \theta] = g(\theta). \quad (28)$$

Then the **total predictive variance** decomposes as

$$\text{Var}(Y_* - g(\hat{\theta})) \approx \nabla_{\theta} g(\hat{\theta})' \mathcal{J}(\hat{\theta})^{-1} \nabla_{\theta} g(\hat{\theta}) + \text{Var}(Y_* | \hat{\theta}), \quad (29)$$

where the first term is due to parameter uncertainty and the second term is intrinsic observation noise.

Assuming asymptotic normality, a $(1 - \alpha)$ **FIM-based prediction interval** for Y_* is

$$Y_* \in g(\hat{\theta}) \pm z_{1-\alpha/2} \sqrt{\nabla_{\theta} g(\hat{\theta})' \mathcal{J}(\hat{\theta})^{-1} \nabla_{\theta} g(\hat{\theta}) + \text{Var}(Y_* | \hat{\theta})}, \quad (30)$$

where $z_{1-\alpha/2}$ is the standard Gaussian quantile. It is worth to note that if Y_* is Gaussian, this interval is exact; otherwise, it is a first-order asymptotic approximation.

Example 1. Consider the general linear regression model:

$$Y = X\theta + \varepsilon, \quad \varepsilon \sim N(0, \sigma^2 I_n), \quad (31)$$

where $\theta \in \mathbb{R}^p$ is unknown and $X \in \mathbb{R}^{n \times p}$ is fixed. For a new design vector $x_* \in \mathbb{R}^p$, the predicted mean is

$$g(\theta) = x_*' \theta. \quad (32)$$

Since g is linear, $\nabla_{\theta}g(\theta) = x_*$. The Fisher Information Matrix is given by

$$\mathcal{J}(\theta) = \frac{1}{\sigma^2} X'X, \quad \mathcal{J}(\hat{\theta})^{-1} = \sigma^2 (X'X)^{-1}. \quad (33)$$

Thus, the parameter-uncertainty contribution to the prediction variance is

$$x_*' \mathcal{J}(\hat{\theta})^{-1} x_* = \sigma^2 x_*' (X'X)^{-1} x_*, \quad (34)$$

whereas the intrinsic noise of the future observation is

$$\text{Var}(Y_* | \hat{\theta}) = \sigma^2. \quad (35)$$

Hence, the total prediction variance is

$$\text{Var}(Y_* - x_*' \hat{\theta}) = \sigma^2 (1 + x_*' (X'X)^{-1} x_*). \quad (36)$$

Finally, the $(1 - \alpha)$ prediction interval for Y_* is

$$Y_* \in x_*' \hat{\theta} \pm z_{1-\alpha/2} \sigma \sqrt{1 + x_*' (X'X)^{-1} x_*}. \quad (37)$$

3 The Profile Likelihood method

In many modelling scenarios, we aim not only to estimate model parameters but also to understand how well they are constrained by the data. The **profile likelihood** is a tool that provides an intuitive way to explore this: it examines how the likelihood changes as we vary one parameter at a time, while re-optimizing the remaining parameters. In other words, it answers the question: “If this parameter were fixed at a certain value, how well could the model still fit the data by adjusting the other parameters?” This approach allows us to quantify uncertainty, detect identifiability issues, and construct confidence intervals for individual parameters.

3.1 PL-based confidence intervals

Consider a model with a log-likelihood function $\ell(\boldsymbol{\theta} | \mathbf{y})$, parametrized by a p -dimensional vector $\boldsymbol{\theta} \in \mathbb{R}^p$. Formally, the profile log-likelihood of $\ell(\boldsymbol{\theta} | \mathbf{y})$ with respect to the j -th parameter θ_j is defined as

$$\tilde{\ell}_j(\beta) = \max_{\boldsymbol{\theta} \in \Theta_j(\beta)} \ell(\boldsymbol{\theta} | \mathbf{y}), \quad (38)$$

where

$$\Theta_j(\beta) = \{\boldsymbol{\theta} \in \Theta \mid \theta_j = \beta\}. \quad (39)$$

In this formulation, θ_j is fixed at β , while the remaining $(p - 1)$ parameters $\boldsymbol{\theta}_{i \neq j}$ are optimized to maximize the likelihood. Given an estimator $\hat{\boldsymbol{\theta}}$, the profile log-likelihood can be used to construct $(1 - \alpha)$ confidence intervals for individual parameters:

$$CI_{1-\alpha} = \left\{ \beta \mid 2(\ell(\hat{\boldsymbol{\theta}}) - \tilde{\ell}_j(\beta)) \leq \chi_{1-\alpha,1}^2 \right\}, \quad (40)$$

where $\chi_{1-\alpha,1}^2$ denotes the $(1 - \alpha)$ quantile of the χ^2 distribution with one degree of freedom [1]. This construction is asymptotically valid under standard regularity conditions [2, 3].

Note 1. A flat profile log-likelihood for θ_j indicates an infinitely wide confidence interval at all levels α , signalling structural non-identifiability [4]. In this case, adjustments in θ_j can be compensated by re-optimizing the other parameters $\boldsymbol{\theta}_{i \neq j}$, leaving the likelihood unchanged. Essentially, the data $\mathbf{y}$ provide no information about θ_j . If the profile log-likelihood $\tilde{\ell}_j(\beta)$ fails to exceed the threshold $\ell(\hat{\boldsymbol{\theta}}) - \frac{1}{2}\chi_{1-\alpha,1}^2$ in at least one direction (left or right of $\beta = \hat{\theta}_j$), θ_j is deemed practically non-identifiable [5]. Here, the data contain some information about θ_j , but the estimate $\hat{\theta}_j$ is highly uncertain due to limited or noisy data. Conversely, if $\tilde{\ell}_j(\beta)$ rises above the threshold on both sides of $\hat{\theta}_j$, the parameter θ_j is structurally identifiable [5]. In this situation, variations in θ_j cannot be offset by adjusting the other parameters, thereby affecting the likelihood.

3.2 PL-based prediction intervals

Although profile likelihood is traditionally used to quantify uncertainty in model parameters, the same likelihood-ratio framework can be extended to construct prediction intervals for future observations. The key idea is to treat the unobserved future response as an additional parameter and to apply likelihood profiling in this augmented space. More formally, for a future observation Y^* at an input t^* , consider the augmented likelihood

$$L(\theta, y^* \mid D) = L(\theta; D) p(y^* \mid \theta), \quad (41)$$

and the profile prediction log-likelihood

$$\ell_p(y^*) = \max_{\theta} [\ell(\theta) + \log p(y^* \mid \theta)]. \quad (42)$$

Then, a $(1 - \alpha)$ **profile-likelihood prediction interval** is

$$\text{PI}_{\text{PL}} = \{y^* : 2[\ell_p(\hat{y}^*) - \ell_p(y^*)] \leq \chi_{1,1-\alpha}^2\}, \quad (43)$$

which is asymptotically valid, that is it has asymptotically $1 - \alpha$ coverage, and accounts for parameter uncertainty and new observation noise within the likelihood framework [2]. However, it is rarely used in practice because it requires re-maximizing the likelihood for each candidate y^* . Therefore, several heuristics exist to approximate profile-likelihood prediction intervals. Such approximations either (i) sample parameter vectors from the profile-likelihood confidence region $\mathcal{C}_\alpha$, generate predictive draws, and take empirical quantiles, or (ii) construct an envelope by taking the minimum and maximum of the model output over $\mathcal{C}_\alpha$ and add measurement noise to it. However, since both procedures are largely heuristic and do not have a basis in classical likelihood theory (see e.g., [6, 7]), they do not guarantee nominal coverage for new observations. These methods are briefly detailed in the following subsections.

Envelope-based PL prediction bands

A practical but informal alternative uses the parameter PL region $\mathcal{C}_\alpha$ to define

$$[\mu_{\min}(t^*), \mu_{\max}(t^*)] = \left[\min_{\theta \in \mathcal{C}_\alpha} f(t^*; \theta), \max_{\theta \in \mathcal{C}_\alpha} f(t^*; \theta) \right] \quad (44)$$

followed by adding observational noise, thus obtaining

$$\text{PI}(Y^* | t^*) \approx [\mu_{\min}(t^*) - k\sigma_\varepsilon, \mu_{\max}(t^*) + k\sigma_\varepsilon], \quad (45)$$

where k is an appropriate quantile (e.g. normal or t -distribution). However, since $\mathcal{C}_\alpha$ is a *set of plausible parameter values* rather than a probability distribution, no probabilistic weighting is applied to the parameters. This means that we are essentially assuming that all values $\theta \in \mathcal{C}_\alpha$ are equally likely to generate a new observation Y^* . Consequently, the resulting interval may under- or overcover the future observation Y^* , and its nominal $1 - \alpha$ coverage is not guaranteed. In contrast, formally calibrated prediction intervals, such as those derived from the likelihood-ratio framework or Bayesian posterior predictive distributions, properly account for both parameter uncertainty and observational noise. Furthermore, while the observational noise term can provide a wider interval that accounts for measurement noise, it does not change the fact that the interval is still based on the profile-likelihood *set* rather than a full probability distribution, so formal coverage guarantees are not ensured.

Sampling from the PL confidence region

Another empirical alternative, but theoretically unjustified, approach that is sometimes suggested is to generate predictive samples by randomly drawing parameter values from the profile-likelihood confidence region. While it may seem intuitive, this procedure again does not correspond to a proper probability model and can lead to uncontrolled coverage. The algorithm of this procedure is defined as follows:

1. Sample many parameter vectors $\theta^{(j)}$ from the profile-likelihood confidence region $\mathcal{C}_\alpha$.
2. For each sampled $\theta^{(j)}$, generate a predictive draw

$$Y^{*(j)} = f(t^*; \theta^{(j)}) + \varepsilon^{*(j)},$$

where $\varepsilon^{*(j)}$ is drawn from the observational noise distribution.

3. Construct an empirical prediction interval by taking the appropriate quantiles (e.g., $\alpha/2$ and $1 - \alpha/2$) of the set $\{Y^{*(j)}\}$.

Since $\mathcal{C}_\alpha$ is a set rather than a probability distribution, this procedure does not guarantee nominal coverage, and the resulting interval can under- or overcover the true future observation Y^* .

4 Conformal Prediction methods

Given a mathematical model $y = f(x)$, linking an input x with an output y , conformal prediction is a model-agnostic framework that produces prediction sets (or intervals) with guaranteed coverage under minimal probabilistic assumptions. More formally, given an input x , the conformal predictor returns a set

$$C(x) \subseteq \mathcal{Y} \quad (46)$$

such that, for a chosen significance level $\alpha \in (0, 1)$,

$$P(Y_{n+1} \in C(X_{n+1})) \geq 1 - \alpha. \quad (47)$$

In particular, the set contains the true output y with probability at least $1 - \alpha$, under the assumption of exchangeability, which is defined as follows.

Definition 1. A sequence of random variables $Z_1, \dots, Z_{n+1}$ is said to be **exchangeable** if the joint distribution is invariant under permutation, that is

$$P(Z_1, \dots, Z_{n+1}) = P(Z_{\pi(1)}, \dots, Z_{\pi(n+1)}), \quad (48)$$

for any permutation π of $\{1, 2, \dots, n+1\}$.

In conformal prediction, each observation $Z_i = (X_i, Y_i)$ represents a feature-label (or input-output) pair. The assumption that $(Z_1, \dots, Z_{n+1})$ are exchangeable means that all data points, including the new test point, are statistically indistinguishable apart from order. It is worth to note that independent and identically distributed (iid) data automatically satisfy exchangeability, but the latter is slightly weaker. Exchangeability ensures that the rank of any point among the others is uniformly distributed, which is key to the $(1 - \alpha)$ coverage guarantee.

Conformal prediction uncertainty quantification relies on measures of nonconformity, which can be defined as follows.

Definition 2. A **nonconformity measure/score** is a function

$$\alpha_i = A(x_i, y_i), \quad (49)$$

of the pair (x_i, y_i) , typically symmetric in the entries, which intuitively quantifies “how strange” the pair (x_i, y_i) looks, compared to the model and the remaining data. For example, in a regression setting, a common choice is

$$\alpha_i = A(x_i, y_i) = |y_i - \hat{f}(x_i)| =: r_i, \quad (50)$$

that is, the absolute residual r_i of the fitted model.

There are different types of conformal prediction methods. The most established ones are called **full conformal prediction** and **split conformal prediction**. These strategies are briefly outlined in the following subsections.

4.1 Full conformal prediction

In full conformal prediction, given observed pairs $(x_1, y_1), \dots, (x_n, y_n)$, a test input x_{n+1} , and a candidate label/output y , the prediction region $C(x_{n+1})$ for x_{n+1} is defined as

$$C(x_{n+1}) = \{y : p(y) > \alpha\}, \quad (51)$$

where $p(y)$ is a p-value-like quantity defined as

$$p(y) = \frac{1}{n+1} \sum_{i=1}^{n+1} \mathbf{1}\{\alpha_i \geq \alpha_{n+1}\}, \quad (52)$$

where α_i are the nonconformity scores computed when (x_{n+1}, y) is temporarily included in the dataset, and the model is refit, and $\mathbf{1}\{\cdot\}$ is the indicator function. Intuitively, a large $p(y)$ indicates that y is a plausible output (since its nonconformity α_{n+1} is not too extreme compared to the other nonconformity scores $\alpha_{i \neq n+1}$), while a small $p(y)$ suggests that y is an unusual output given the input x_{n+1} . The algorithm of the full conformal prediction can be summarised as follows:

Given training data $(x_1, y_1), \dots, (x_n, y_n)$ and a test input x_{n+1} :

1. For each possible candidate label $y \in \mathcal{Y}$:

- (a) Form the augmented dataset $\mathcal{D}' = \{(x_1, y_1), \dots, (x_n, y_n), (x_{n+1}, y)\}$.
- (b) Fit the model f' on $\mathcal{D}'$.
- (c) Compute nonconformity scores

$$\alpha_i = A(x_i, y_i), \quad i = 1, \dots, n+1.$$

- (d) Compute

$$p(y) = \frac{1}{n+1} \sum_{i=1}^{n+1} \mathbf{1}\{\alpha_i \geq \alpha_{n+1}\}.$$

2. The prediction set is

$$C(x_{n+1}) = \{y : p(y) > \alpha\}.$$

Proposition 2. *Assuming exchangeability, given a significance level $\alpha \in (0, 1)$, the full conformal prediction set given by Eq. (66) achieves coverage of $(1 - \alpha)$.*

Proof. (sketch of)

Let $Z_i = (X_i, Y_i)$. Since $(Z_1, \dots, Z_{n+1})$ are exchangeable, so are their nonconformity scores $(\alpha_1, \dots, \alpha_{n+1})$ (the nonconformity function is a measurable mapping, and therefore it preserves distributional equality). This means that the ranks $R_1 = r(\alpha_1), \dots, R_{n+1} = r(\alpha_{n+1})$ of the nonconformity scores $(\alpha_1, \dots, \alpha_{n+1})$ are uniformly distributed. In fact,

$$r(\alpha_1, \dots, \alpha_{n+1}) \stackrel{d}{=} r(\alpha_{\pi(1)}, \dots, \alpha_{\pi(n+1)}), \quad (53)$$

for any permutation $\pi(\cdot)$. However,

$$r(\alpha_{\pi(1)}, \dots, \alpha_{\pi(n+1)}) = \pi(r(\alpha_1, \dots, \alpha_{n+1})). \quad (54)$$

holds for any permutation $\pi(\cdot)$. Therefore, the distribution d of the rank vector $r(\alpha_1, \dots, \alpha_{n+1})$ must be invariant under all permutations of its coordinates. The only distribution d that allows for this is the uniform distribution. This means that

$$P(R_{n+1} \leq k) = \frac{k}{n+1}. \quad (55)$$

Now,

$$p(y_{n+1}) = \frac{1}{n+1} \sum_{i=1}^{n+1} \mathbf{1}\{\alpha_i \geq \alpha_{n+1}\} = \frac{R_{n+1}}{n+1}. \quad (56)$$

Therefore,

$$P(p(y_{n+1}) \leq \alpha) = P\left(\frac{R_{n+1}}{n+1} \leq \alpha\right) = P(R_{n+1} \leq \alpha(n+1)) = \frac{\lfloor \alpha(n+1) \rfloor}{n+1} \leq \frac{\alpha(n+1)}{n+1} = \alpha. \quad (57)$$

Thus,

$$P(y_{n+1} \in C(x_{n+1})) = P(p(y_{n+1}) > \alpha) = 1 - P(p(y_{n+1}) \leq \alpha) \geq 1 - \alpha, \quad (58)$$

which completes the proof. $\square$

Remarks 1. *It is worth to note that although the full conformal prediction provides exact $(1 - \alpha)$ coverage, this approach is computationally expensive because the model must be refit for each candidate y . For this reason this method is mainly used for theoretical understanding or simple models.*

4.2 Split Conformal Prediction

Idea: To avoid refitting the model many times, the data are split into a **training set** $\mathcal{D}_{\text{train}}$ and a **calibration set** $\mathcal{D}_{\text{cal}}$. The model is first trained once on the training set $\mathcal{D}_{\text{train}}$, and subsequently the calibration set $\mathcal{D}_{\text{cal}}$ is used to estimate the distribution of residuals $r_i = |y_i - \hat{f}(x_i)|$, $(x_i, y_i) \in \mathcal{D}_{\text{cal}}$, where $\hat{f}$ indicate a model fitted under the training set. Finally, the **split prediction set** for a test input x_{n+1} is defined as

$$C(x_{n+1}) = \left\{ y \in \mathcal{Y} : |y - \hat{f}(x_{n+1})| \leq q_{1-\alpha} \right\} \quad (59)$$

where $q_{1-\alpha}$ is the $(1-\alpha)(1+1/m)$ order quantile of the calibration residuals $r_i = |y_i - \hat{f}(x_i)|$, $(x_i, y_i) \in \mathcal{D}_{\text{cal}}$, that is

$$q_{1-\alpha} = \text{quantile}_{(1-\alpha)(1+1/m)}(\{r_i\}_{i=1}^m). \quad (60)$$

Given the data $(x_1, y_1), \dots, (x_n, y_n)$ and a test input x_{n+1} , split conformal prediction algorithm can be summarised as follows:

1. Split the data into $\mathcal{D}_{\text{train}}$, $\mathcal{D}_{\text{cal}}$, $|\mathcal{D}_{\text{cal}}| = m$.
2. Fit f on $\mathcal{D}_{\text{train}}$, obtaining $\hat{f}$.
3. Compute calibration residuals: $r_i = |y_i - \hat{f}(x_i)|$, $(x_i, y_i) \in \mathcal{D}_{\text{cal}}$.
4. Let $q_{1-\alpha}$ be the $\lceil (1-\alpha)(m+1) \rceil$ -th smallest value in $\{r_i\}_{i=1}^m$. This is approximately achieved by setting

$$q_{1-\alpha} = \text{quantile}_{(1-\alpha)(1+1/m)}(\{r_i\}_{i=1}^m).$$

5. For a new input x_{n+1} , define the prediction interval

$$C(x_{n+1}) = [\hat{f}(x_{n+1}) - q_{1-\alpha}, \hat{f}(x_{n+1}) + q_{1-\alpha}].$$

Proposition 3. *Under the assumption that $(Z_1, \dots, Z_n) = ((X_1, Y_1), \dots, (X_n, Y_n))$ are exchangeable, the split conformal prediction set $C(x_{n+1})$ of Eq. (59) satisfies*

$$P(y_{n+1} \in C(x_{n+1})) \geq 1 - \alpha, \quad (61)$$

that is the split conformal prediction construction guarantees finite-sample coverage under exchangeability.

Proof. Let (x_{n+1}, y_{n+1}) a new point. By assumption, the calibration data $(x_i, y_i) \in \mathcal{D}_{\text{cal}}$ and the new point (x_{n+1}, y_{n+1}) are exchangeable. This means that also their scores

$$\alpha_i := |y_i - \hat{f}(x_i)|, \quad \alpha_{n+1} := |y_{n+1} - \hat{f}(x_{n+1})| \quad (62)$$

are exchangeable. Because of exchangeability, the ranks

$$R_i := r(\alpha_i), \quad i = 1, \dots, n+1, \quad (63)$$

are also exchangeable. This means that

$$\begin{aligned} P(y_{n+1} \in C(x_{n+1})) &= P(\alpha_{n+1} \leq q_{1-\alpha}) \\ &= P(r(\alpha_{n+1}) \leq r(q_{1-\alpha})) \\ &= \frac{r(q_{1-\alpha})}{m+1} \\ &= \frac{\lceil (1-\alpha)(m+1) \rceil}{m+1} \\ &\geq \frac{(1-\alpha)(m+1)}{m+1} = (1-\alpha). \end{aligned} \quad (64)$$

□

Remarks 2. *It is worth to note that the split conformal prediction guarantees finite-sample coverage under exchangeability, whilst requiring only one model fit. This approach usually provides slightly more conservative prediction sets (intervals a bit wider) compared to the full conformal approach.*

4.3 Jackknife conformal prediction

Given observed pairs $\mathcal{D} = \{(x_1, y_1), \dots, (x_n, y_n)\}$, a test input x_{n+1} , and a candidate label/output y , the Jackknife conformal prediction builds a prediction region $C(x_{n+1})$ for x_{n+1} by first fitting the model f on every leave-one-out subsample $\mathcal{D} \setminus \{(x_i, y_i)\}$, $\forall i = 1, \dots, n$, providing the **jackknife residuals**

$$\alpha_i = |y_i - \hat{f}_{-i}(x_i)|. \quad (65)$$

Then, the prediction region $C(x_{n+1})$ for x_{n+1} is defined as

$$C(x_{n+1}) = \{y \in \mathcal{Y} : |y - \hat{f}(x)| \leq q_{1-\alpha}\}, \quad (66)$$

where

$$q_{1-\alpha} = \text{quantile}_{1-\alpha}(\alpha_1, \dots, \alpha_n) \quad (67)$$

where α_i are the jackknife nonconformity scores as defined above. It can be shown that this strategy can provide prediction regions having only approximate coverage of $(1 - \alpha)$. An improved variant of this method, called the Jackknife+, has been proposed to overcome this limitation [8]. The Jackknife+ algorithm is summarised as follows:

1. For each i :

- Train the model f on every leave-one-out subsample $\mathcal{D} \setminus \{(x_i, y_i)\}$, $\forall i = 1, \dots, n$
- Compute predictions for the test input as

$$\hat{y}_{-i}(x_{n+1}) = \hat{f}_{-i}(x_{n+1}) \quad (68)$$

- Calculate the absolute residuals

$$r_i = |y_i - \hat{f}_{-i}(x_i)| \quad (69)$$

2. For each training point (x_i, y_i) define the interval

$$I_i = [\min\{\hat{y}_{-i}(x_{n+1}), y_i\} - r_i, \max\{\hat{y}_{-i}(x_{n+1}), y_i\} + r_i]. \quad (70)$$

3. Given a level $\alpha \in (0, 1)$, the Jackknife+ prediction interval for the test input x_{n+1} is defined as

$$C_{JK+}(x_{n+1}) = [q_{1-\alpha}(\{I_i^l\}_{i=1}^n), q_\alpha(\{I_i^u\}_{i=1}^n)] \quad (71)$$

$$I_i^l = \min\{\hat{y}_{-i}(x_{n+1}), y_i\} - r_i, \quad I_i^u = \max\{\hat{y}_{-i}(x_{n+1}), y_i\} + r_i.$$

It can be shown that under exchangeability this construction ensures that

$$P(y_{n+1} \in C_{JK+}(x_{n+1})) \geq 1 - \alpha, \quad (72)$$

for any finite sample size n [8].

5 Discrete Markov chains

Definition 3. A *discrete Markov chain of first order* is a stochastic process $\{X_t\}_t$, $t = 1, 2, 3, \dots$ with the discrete state space S , whose **transition probabilities** fulfill the **Markov property**

$$P(X_t = x_t | X_{t-1} = x_{t-1}, \dots, X_1 = x_1) = P(X_t = x_t | X_{t-1} = x_{t-1}), \quad (73)$$

for all states $x_1, \dots, x_t \in S$. A discrete Markov chain is called **homogeneous** if the transition probabilities do not depend on t , that is

$$T(x, y) := P(X_t = y | X_{t-1} = x) = P(X_2 = y | X_1 = x), \quad (74)$$

which is then the 1-step **transition kernel** of the homogeneous (discrete) 1st-order Markov chain.

In the following, we assume that the state space S is given by a finite set. Furthermore, we assume, without loss of generality, that the state space is given by $S = \{1, 2, 3, \dots, k\}$, where k is the number of possible states, or the **cardinality** of S . This means that the transition kernel $T(\cdot, \cdot)$ can be represented as a $k \times k$ matrix, called the **transition matrix**, whose (i, j) -th element is defined as

$$T_{ij} := T(i, j) = P(X_{t+1} = j | X_t = i) = P(X_2 = j | X_1 = i). \quad (75)$$

The **initial distribution** $P(X_1 = x)$, $x \in S$, and the transition matrix T together fully define the distribution of the homogeneous Markov chain.

Definition 4. A **stochastic matrix** is a square matrix whose row or column sums are equal to one, and whose elements take values between zero and one.

Note 2. It is worth noting that the transition matrices of discrete Markov chains (with finite state space) are stochastic matrices, as each row-sum is equal to 1.

Example 2. Let's consider a Markov chain with $k = 4$ states, $S = \{1, 2, 3, 4\}$, initial distribution $P(X_1 = i) = 0.25$, $i = 1, \dots, 4$, and transition matrix

$$T = \begin{bmatrix} 0.5 & 0.45 & 0.05 & 0 \\ 0.48 & 0.5 & 0 & 0.02 \\ 0.01 & 0 & 0.99 & 0 \\ 0 & 0.001 & 0 & 0.999 \end{bmatrix} \quad (76)$$

Then, the distribution at the next two time points can be computed iteratively as follows

$$\begin{aligned} P(X_2 = j) &= \sum_{i=1}^4 P(X_1 = i) T_{ij} \\ P(X_3 = j) &= \sum_{i=1}^4 P(X_2 = i) T_{ij} \\ &= \sum_{i=1}^4 \left(\sum_{k=1}^4 P(X_1 = k) T_{ki} \right) T_{ij} \\ &= \sum_{k=1}^4 P(X_1 = k) \sum_{i=1}^4 T_{ki} T_{ij} \\ &= \sum_{k=1}^4 P(X_1 = k) (TT)_{kj} \\ &= \sum_{k=1}^4 P(X_1 = k) T_{kj}^2 \\ &= \sum_{i=1}^4 P(X_1 = i) T_{ij}^2. \end{aligned} \quad (77)$$

This result can be generalised as follows

Proposition 4. For discrete homogeneous Markov chains with finite state space $S = \{1, 2, \dots, k\}$, an initial distribution $P(X_1 = i) = p_{1i}$, $i = 1, \dots, k$, and k -by- k transition matrix T , we have

$$P(X_t = j) = \sum_{i=1}^k p_{1i} T_{ij}^{t-1}, \quad (78)$$

and thus

$$(p_{t1}, \dots, p_{tk}) = (p_{11}, \dots, p_{1k}) T^{t-1} \\ \mathbf{p}_t = \mathbf{p}_1 T^{t-1}, \quad (79)$$

where

$$\mathbf{p}_t = (p_{t1}, \dots, p_{tk}). \quad (80)$$

That is, the $(t-1)$ -step transition probabilities $\mathbf{p}_t$ at time t can be obtained by multiplying the initial distribution $\mathbf{p}_1$ with T^{t-1} , the transition matrix raised to the $(t-1)$ power.

Example 3. Compute the values of $\mathbf{p}_t$, for $t = 1, 2, 3, 4, 8, 16, 32, 64, 128, 256, 512, 2024, 4048, 9096$, for the Markov chain of previous example.

Solution:

```
T <- matrix(data = c(0.5,0.45,0.05,0,
                    0.48,0.5,0,0.02,
                    0.01,0,0.99,0,
                    0,0.001,0,0.999),
            byrow = TRUE,
            nrow = 4,
            ncol = 4)

p0 <- rep(0.25,4)

tps <- c(1,2,3,4,8,16,32,64,128,256,512,2024,4048,9096)

res <- t(sapply(tps, function(k){
  Tk <- T
  if(k > 1){
    for (i in 1:(k-1)) {
      Tk <- Tk %*% T
    }
  }
  return(p0 %*% Tk)
}))
rownames(res) <- tps

xtable::xtable(res)
```

Note that by repeating the same exercise with a different initial probability distribution $\mathbf{p}_1$ will lead to the same distribution at the final iteration. We will refer later to this distribution as the stationary distribution of the Markov chain. $\square$

A concrete trajectory (realization) of a given Markov chain can be generated via the following two-step iterative procedure.

1. **(Initialization):** Draw a realization x_1 of the random variable X_1 from the initial distribution $\mathbf{p}_1$.
2. **(Iteration):** For $t = 2, \dots, T$, given the realization $x_{t-1} = i$ of X_{t-1} , draw a realization of the next random variable X_t from $P(X_t | X_{t-1} = i)$. This conditional distribution is given by

$$P(X_t = j | X_{t-1} = i) = T_{ij}, \quad (81)$$

that is the i -th row of the transition matrix T .

	$i = 1$	$i = 2$	$i = 3$	$i = 4$
p_1	0.2475	0.2377	0.2600	0.2547
p_2	0.2405	0.2305	0.2698	0.2593
p_3	0.2336	0.2237	0.2791	0.2636
p_4	0.2270	0.2172	0.2880	0.2678
p_8	0.2032	0.1939	0.3195	0.2834
p_{16}	0.1663	0.1577	0.3666	0.3093
p_{32}	0.1214	0.1137	0.4177	0.3472
p_{64}	0.0859	0.0792	0.4399	0.3950
p_{128}	0.0699	0.0640	0.4093	0.4568
p_{256}	0.0596	0.0549	0.3420	0.5435
p_{512}	0.0486	0.0451	0.2634	0.6429
p_{2024}	0.0390	0.0366	0.1954	0.7290
p_{4048}	0.0389	0.0365	0.1946	0.7299
p_{9096}	0.0389	0.0365	0.1946	0.7299

After k iterations, it is also possible to compute the relative frequencies h_{ki} , defining how often the J states have been realized.

Example 4. Let's consider again a Markov chain with $k = 4$ states, $S = \{1, 2, 3, 4\}$, initial distribution $P(X_1 = i) = 0.25$, $i = 1, \dots, 4$, and transition matrix

$$T = \begin{bmatrix} 0.5 & 0.45 & 0.05 & 0 \\ 0.48 & 0.5 & 0 & 0.02 \\ 0.01 & 0 & 0.99 & 0 \\ 0 & 0.001 & 0 & 0.999 \end{bmatrix}. \quad (82)$$

Simulate a concrete trajectory (realization) of this Markov chain. Generate realizations of the first 10000 random variables, and report the relative frequencies h_{ki} , every 1000 steps.

Solution:

```
nIT <- 10000
x <- rep(NA, nIT)
x_freq <- matrix(data = 0, nrow = 10, ncol = 4)
colnames(x_freq) <- 1:4
x[1] <- sample(x = 1:4, size = 1, prob = p0)
for (k in 2:nIT) {
  x[k] <- sample(x = 1:4, size = 1, prob = T[x[k-1],])

  if(k %% 1000 == 0){
    x_freq[k/1000, names(table(x[1:k]))] <- table(x[1:k])/k
  }
}
xtable::xtable(x_freq, digits = 4)
```

Note that the values of last row of this table are comparable to those of the distribution at last iteration obtained in the previous exercise. $\square$

5.1 Stationary distribution of a discrete Markov chain

Any discrete distribution π over a set $S = \{1, \dots, k\}$ is fully specified by its discrete density function. Let p_i be the probability of the realization of the value $i \in \{1, \dots, k\}$. We further assume that $p_i > 0$ for $i = 1, \dots, k$. That is, we assume that each $i \in S$ has a positive probability. Note that each $i \in S$ with $p_i = 0$ can be simply excluded from S . Then, the discrete distribution π can be identified with the row vector $\pi = (p_1, \dots, p_k)$.

	1	2	3	4
1	0.0050	0.0140	0.4150	0.5660
2	0.0025	0.0070	0.2075	0.7830
3	0.0077	0.0130	0.1417	0.8377
4	0.0150	0.0215	0.1095	0.8540
5	0.0204	0.0266	0.1292	0.8238
6	0.0262	0.0293	0.2225	0.7220
7	0.0296	0.0313	0.2293	0.7099
8	0.0348	0.0362	0.2160	0.7130
9	0.0309	0.0322	0.1920	0.7449
10	0.0319	0.0319	0.1728	0.7634

Definition 5. A discrete distribution $\pi = (p_1, \dots, p_k)$ over a set $S = \{1, \dots, k\}$ is called a **stationary distribution** of a discrete Markov chain with transition matrix T if

$$\pi T = \pi. \quad (83)$$

Note 3. The stationary distribution of a Markov chain does not depend on the initial distribution, and it is fully specified by the transition matrix T .

5.2 Aperiodic Markov chains

Definition 6. A discrete Markov chain defined over a finite state space S is called **periodic** if it can reach some states of the state space S only in a certain order, otherwise the Markov chain is called **aperiodic**.

Note 4. A discrete Markov chain with finite state space S is aperiodic if for each state $i \in S$ there is a natural number $n_i \in \mathbb{N}$ such that

$$P(X_t = i | X_1 = i) > 0 \quad \forall t \geq n_i, \quad (84)$$

that is, for every state $i \in S$ the return to this state can occur in irregular time intervals, where the return may take place not earlier than after n_i steps (or transitions). This means that a Markov chain with finite state space S is periodic if there is at least one state $i \in S$ and a natural number $n_i \geq 2$, such that

$$\begin{aligned} P(X_t = i | X_1 = i) &> 0 \quad \text{for } t = n_i, 2n_i, 3n_i, \dots \\ P(X_t = i | X_1 = i) &= 0 \quad \text{for all other } t. \end{aligned} \quad (85)$$

5.3 Irreducible and ergodic Markov chains

Definition 7. A discrete Markov chain with finite state space S is called **irreducible** if every state can be reached from every other state in a finite number of steps with positive probability, i.e. if

$$\exists t > 0 \quad \text{such that } P(X_t = j | X_1 = i) > 0 \quad \forall i, j \in S. \quad (86)$$

Definition 8. A discrete Markov chain with finite state space S is called **ergodic** if it is aperiodic and irreducible.

Proposition 5. For a discrete ergodic Markov chain with finite state space S and transition matrix T , there is a unique stationary distribution π . The stationary distribution π is fully determined by the transition matrix T and it holds that

$$\pi = \mathbf{p}_1 T^t \quad \text{for } t \rightarrow \infty, \quad (87)$$

where $\mathbf{p}_1$ is an arbitrary initial distribution.

Note 5. The previous propositions implies that each discrete ergodic Markov chain with finite state space S and transition matrix T “converges” to a unique stationary distribution π , and this stationary distribution depends only on the transition matrix T .

Proposition 6. Given a discrete ergodic Markov chain with finite state space $S = \{1, \dots, k\}$ and transition matrix T , if a distribution $\pi = (p_1, \dots, p_k)$, with $p_i > 0$, fulfills for all $i, j \in S$ the so-called **equation of detailed balance**

$$\frac{T_{ij}}{T_{ji}} = \frac{p_j}{p_i}, \quad (88)$$

then the distribution π is the stationary distribution of this Markov chain.

Proof. (sketch of)

$$\begin{aligned} \frac{T_{ij}}{T_{ji}} &= \frac{p_j}{p_i} \\ T_{ij} \cdot p_i &= T_{ji} \cdot p_j \\ \sum_{i=1}^k T_{ij} \cdot p_i &= \sum_{i=1}^k T_{ji} \cdot p_j \\ \sum_{i=1}^k T_{ij} \cdot p_i &= \sum_{i=1}^k P(X_2 = i | X_1 = j) \cdot p_j \\ \sum_{i=1}^k T_{ij} \cdot p_i &= p_j \underbrace{\sum_{i=1}^k P(X_2 = i | X_1 = j)}_{=1} \\ \sum_{i=1}^k T_{ij} \cdot p_i &= p_j \\ \pi T &= \pi. \end{aligned} \quad (89)$$

□

Note 6. The equation (88) of detailed balance is always satisfied for $i = j$, as we have for $T_{ii} > 0$

$$1 = \frac{T_{ii}}{T_{ii}} = \frac{p_i}{p_i}, \quad (90)$$

and also note that, by definition, the equation (88) of detailed balance does not have to be satisfied if $T_{ii} = 0$.

6 Markov Chain Monte Carlo (MCMC)

6.1 Motivation

Given a discrete distribution $\pi = (p_1, \dots, p_k)$, construct an ergodic discrete Markov chain with state space $S = \{1, \dots, k\}$ and transition matrix T , so that the equation of detailed balance is fulfilled. The distribution π is the stationary distribution of this Markov chain. That is, the so constructed Markov chain “converges” to π in the sense that if this Markov chain is simulated for “long enough”, it generates a random sample from the distribution π .

6.2 Metropolis-Hastings method

With the so-called “Metropolis-Hastings Markov Chain Monte Carlo” simulation method such a Markov chain can be constructed as follows:

For iteration $t = 1, 2, 3, \dots$

- Given the state $x_t = i$ of the Markov chain, a new state $j \in S$ is proposed with a **proposal probability** $Q(i, j)$.
- The new proposed state j is either accepted with **acceptance probability** $A(i, j)$, or rejected with the counter probability $1 - A(i, j)$.

- If the new state is accepted, set $x_{t+1} = j$, otherwise leave the state unchanged, that is, set $x_{t+1} = i$.

Let's consider the **transition kernel** defined as

$$T(i, j) = T_{ij}, \quad (91)$$

where T_{ij} is the ij -th entry of the transition matrix T . The transition probabilities $T(i, j)$ from state i to state j have to be constructed with the so called **Metropolis-Hastings** method, that is

$$T(i, j) = \begin{cases} Q(i, j)A(i, j) & \text{if } j \neq i \\ Q(i, i)A(i, i) + 1 - \sum_{k \neq i} Q(i, k)A(i, k) & \text{if } j = i. \end{cases} \quad (92)$$

The diagonal elements of T are given by

$$T(i, i) = Q(i, i)A(i, i) + 1 - \sum_{j \neq i} Q(i, j)A(i, j), \quad (93)$$

although transitions from i to i are usually not explicitly proposed and, in fact, $Q(i, i)$ are often set to zero, so that $T(i, i)$ reduces to

$$T(i, i) = 1 - \sum_{j \neq i} Q(i, j)A(i, j). \quad (94)$$

In this case $T(i, i)$ is then simply the probability that a new proposed state $j \neq i$ is not accepted, so that the state of the Markov chain remains unchanged.

Note 7. In the Metropolis-Hastings MCMC method:

- The proposal probabilities $Q(i, j)$ are chosen arbitrarily, as long as the following two conditions are fulfilled:
 1. each state can be reached from each state in a finite number of steps.
 2. for $Q(i, j) > 0$ it follows that $Q(j, i) > 0$ too.
- Given the proposal probabilities $Q(i, j)$, the acceptance probabilities $A(i, j)$ have to be chosen so that the Equation (88) of detailed balance is satisfied.

Proposition 7. Consider a discrete distribution $\pi = (p(1), \dots, p(k))$ over the state space $S = \{1, \dots, k\}$. In the Metropolis-Hastings MCMC method, given the proposal probabilities $Q(i, j)$, the acceptance probabilities $A(i, j)$ defined as

$$A(i, j) = \min \left\{ 1, \frac{p(j)}{p(i)} \frac{Q(j, i)}{Q(i, j)} \right\} \quad (95)$$

satisfy the Equation (88) of detailed balance, that is

$$\frac{T_{ij}}{T_{ji}} = \frac{p(j)}{p(i)}. \quad (96)$$

This means that, any Markov chain simulated for “long enough” by the Metropolis-Hastings method, with the proposal probabilities as defined above, it will generate a random sample from the distribution π .

Proof. (sketch of) We want to prove that

$$\frac{T_{ij}}{T_{ji}} = \frac{Q(i, j)A(i, j)}{Q(j, i)A(j, i)} = \frac{p(j)}{p(i)}. \quad (97)$$

By definition of the acceptance probabilities, we have

$$A(i, j) = \min \left\{ 1, \frac{p(j)}{p(i)} \frac{Q(j, i)}{Q(i, j)} \right\} \quad \text{and} \quad A(j, i) = \min \left\{ 1, \frac{p(i)}{p(j)} \frac{Q(i, j)}{Q(j, i)} \right\} \quad (98)$$

We can distinguish two cases:

- $A(i, j) < 1$:

$$\begin{aligned} \min \left\{ 1, \frac{p(j)}{p(i)} \frac{Q(j, i)}{Q(i, j)} \right\} &< 1 \\ \frac{p(j)}{p(i)} \frac{Q(j, i)}{Q(i, j)} &< 1 \\ \frac{p(i)}{p(j)} \frac{Q(i, j)}{Q(j, i)} &> 1 \\ A(j, i) &= 1 \end{aligned} \tag{99}$$

This means that

$$\frac{A(i, j)}{A(j, i)} = \frac{A(i, j)}{1} = \frac{p(j)}{p(i)} \frac{Q(j, i)}{Q(i, j)}. \tag{100}$$

- $A(i, j) = 1$ implies (it is an “if and only if”) $A(j, i) < 1$. This means that

$$\frac{A(i, j)}{A(j, i)} = \frac{1}{A(j, i)} = \frac{1}{\frac{p(i)}{p(j)} \frac{Q(i, j)}{Q(j, i)}} = \frac{p(j)}{p(i)} \frac{Q(j, i)}{Q(i, j)}. \tag{101}$$

So, in either case

$$\frac{A(i, j)}{A(j, i)} = \frac{p(j)}{p(i)} \frac{Q(j, i)}{Q(i, j)}, \tag{102}$$

from which it follows that

$$\frac{T_{ij}}{T_{ji}} = \frac{Q(i, j)A(i, j)}{Q(j, i)A(j, i)} = \frac{Q(i, j)}{Q(j, i)} \frac{p(j)}{p(i)} \frac{Q(j, i)}{Q(i, j)} = \frac{p(j)}{p(i)}. \tag{103}$$

□

Remark 1. *It is worth to note that*

1. *Since the acceptance probabilities $A(i, j) = \min \left\{ 1, \frac{p(j)}{p(i)} \frac{Q(j, i)}{Q(i, j)} \right\}$ in the Metropolis-Hastings MCMC method depend only on the ratios of (discrete) densities, it is sufficient if this ratio can be computed. Also, for the computation of this (density) ratio, the normalization constant does not need to be known.*
2. *To accept a new state with a certain acceptance probability $A(i, j)$, in practice it means that a random number u has to be drawn from a continuous uniform distribution over the interval $[0, 1]$, that is $u \sim U(0, 1)$. Then*
 - *if $u \leq A(i, j)$, the new proposed state is accepted,*
 - *if $u > A(i, j)$, the new proposed state is rejected.*
3. *The Metropolis-Hastings MCMC method can also be applied if the state space S is (countable or uncountable) infinite. Here we have restricted on finite spaces for simplicity. Two examples of MCMC simulations for (i) a countable infinite and (ii) an uncountable infinite state space S are provided in the following.*

Example 5. *(MCMC sampler with countable infinite S)*

Given a Poisson distribution with parameter λ and the discrete density function

$$p(i) = \frac{\lambda^i}{i!} e^{-\lambda}, \quad i = 0, 1, 2, 3, \dots, \tag{104}$$

construct a Markov chain with state space $S = \mathbb{N}_0$, which converges to the distribution

$$\pi = (p(0), p(1), \dots) \tag{105}$$

as stationary distribution.

Solution: For the proposal distribution, we set

$$Q(0, j) = \begin{cases} 1 & j = 1 \\ 0 & \text{otherwise} \end{cases} \quad (106)$$

for $i = 0$, and

$$Q(i, j) = \begin{cases} 0.5 & j = i + 1 \\ 0.5 & j = i - 1 \\ 0 & \text{otherwise} \end{cases} \quad (107)$$

for $i > 0$. Note that such a proposal distribution satisfies the properties listed in Note 7.

Given the proposal probabilities $Q(i, j)$, the acceptance probabilities $A(i, j)$ have to be chosen so that the Equation (88) is satisfied. From Proposition 7, the equation of detailed balance is satisfied by the following acceptance probabilities

$$\begin{aligned} A(i, j) &= \min \left\{ 1, \frac{p(j)}{p(i)} \frac{Q(j, i)}{Q(i, j)} \right\} \\ &= \min \left\{ 1, \frac{\frac{\lambda^j}{j!} e^{-\lambda}}{\frac{\lambda^i}{i!} e^{-\lambda}} HR(i, j) \right\} \\ &= \min \left\{ 1, \lambda^{j-i} \frac{i!}{j!} HR(i, j) \right\} \end{aligned} \quad (108)$$

where

$$HR(i, j) = \frac{Q(j, i)}{Q(i, j)} = \begin{cases} HR(0, 1) = 0.5 \\ HR(1, 0) = 2 \\ HR(i, j) = 1 & \text{otherwise} \end{cases} \quad (109)$$

```
nSim <- 20000
x <- rep(NA, nSim)
x[1] <- 5

lambda <- 5

for (k in 2:nSim) {
  ## propose new state x_new from Q(i,j):
  if(x[k-1] == 0){
    x_new <- 1
  }else{
    p <- rbinom(n = 1, size = 1, prob = .5)
    if(p == 0){
      x_new <- x[k-1] - 1
    }else{
      x_new <- x[k-1] + 1
    }
  }
}

## define Hastings ratio:
if(x[k-1] == 0 & x_new == 1){
  HR <- 0.5
}else{
  if(x[k-1] == 1 & x_new == 2){
    HR <- 2
  }else{
    HR <- 1
  }
}
```

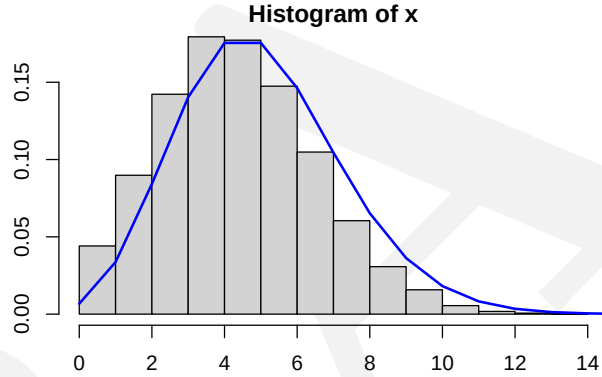
```

## accept or reject x_new according to A(i,j):
Aij <- min(1,
           lambda^(x_new - x[k-1])*factorial(x[k-1])/factorial(x_new)*HR)
u <- runif(n = 1,
           min = 0,
           max = 1)

if(u <= Aij){
  x[k] <- x_new
}else{
  x[k] <- x[k-1]
}
}

hist(x,
     prob = TRUE,
     breaks = sort(unique(x)))
lines(0:20,
      dpois(x = 0:20, lambda = lambda))

```



□

Example 6. (MCMC sampler with uncountable infinite S) Given the standard normal distribution with density

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2} = \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}x^2} \propto_x e^{-\frac{1}{2}x^2}, \quad (110)$$

construct a Markov chain with state space $S = \mathbb{R}$, which converges to the standard normal distribution as the stationary distribution.

Solution: Given the current state x , we propose a new state y such that

$$y = x + u, \quad u \sim U\left(\left[-\frac{1}{2}, \frac{1}{2}\right]\right). \quad (111)$$

This means that the proposal probabilities are defined as

$$Q(x, y) = \begin{cases} 1 & y \in [x - \frac{1}{2}, x + \frac{1}{2}] \\ 0 & \text{otherwise} \end{cases} \quad (112)$$

Note that such a proposal distribution satisfies the properties listed in Note 7.

Given the proposal probabilities $Q(x, y)$, the acceptance probabilities $A(x, y)$ have to be chosen so that the Equation (88) is satisfied. From Proposition 7, the equation of detailed balance is satisfied by the

following acceptance probabilities

$$\begin{aligned}
 A(x, y) &= \min \left\{ 1, \frac{p(y) Q(y, x)}{p(x) Q(x, y)} \right\} \\
 &= \min \left\{ 1, \frac{e^{-\frac{1}{2}y^2}}{e^{-\frac{1}{2}x^2}} \frac{1}{1} \right\} \\
 &= \min \left\{ 1, e^{\frac{1}{2}(x^2 - y^2)} \right\}.
 \end{aligned} \tag{113}$$

```

nSim <- 20000
x <- rep(NA, nSim)
x[1] <- 0

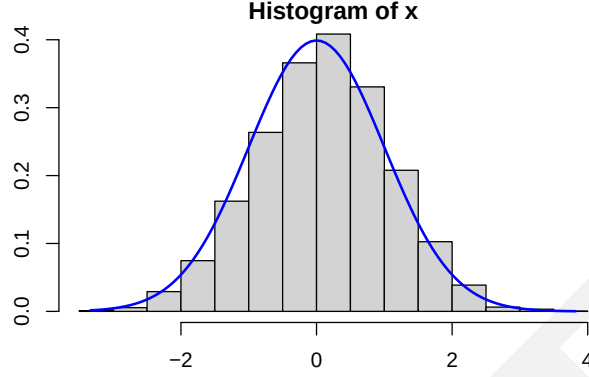
for (k in 2:nSim) {
  ## propose new state x_new from Q(i,j):
  x_new <- x[k-1] + runif(1, -0.5, 0.5)

  ## accept or reject x_new according to A(i,j):
  Aij <- min(1, exp(0.5*(x[k-1]^2 - x_new^2)))
  u <- runif(n = 1,
             min = 0,
             max = 1)

  if(u <= Aij){
    x[k] <- x_new
  }else{
    x[k] <- x[k-1]
  }
}

pdf("./standard-normal-histogram.pdf", width = 5, height = 3)
par(mar = c(2,2,1,1))
hist(x,
      prob = TRUE)
lines(seq(min(x),
          max(x),
          length.out = 100),
      dnorm(x = seq(min(x),
                    max(x),
                    length.out = 100),
            mean = 0,
            sd = 1),
      col = 'blue',
      lwd = 2)
dev.off()

```



□

7 Bayesian inference

In Bayesian statistics, the unknown parameters θ are considered as random variable following certain **prior distributions** that are imposed on them. The prior(s) should reflect the pre-knowledge (prior belief) about the parameters θ . The prior distribution has a density $p(\theta)$. Then, given observed data $\mathbf{y} = (y_1, \dots, y_n)'$, the density $p(\mathbf{y}|\theta)$ is the **likelihood**, describing how “likely” the data $\mathbf{y}$ would be if θ was the true vector parameter. The goal of a Bayesian analysis is to compute the **posterior distribution** $p(\theta|\mathbf{y})$, or at least to generate some samples from it, so as to approximate its properties/features. From Bayes theorem, we know that

$$p(\theta|\mathbf{y}) = \frac{p(\mathbf{y}|\theta)p(\theta)}{p(\mathbf{y})}. \quad (114)$$

Furthermore, since the marginal likelihood $p(\mathbf{y})$ does not depend on θ , the posterior density is proportional, as a function of only the parameters θ , to the likelihood times the prior density, that is

$$p(\theta|\mathbf{y}) \propto p(\mathbf{y}|\theta)p(\theta). \quad (115)$$

A prior is called **conjugate** if the posterior distribution falls in the same family type as the prior distribution. If the posterior distribution density has not the shape of the density of any known distribution, Markov Chain Monte Carlo (MCMC) simulations are used to generate samples from the posterior distribution. In the next subsection, a short review of MCMC simulations is given, for the case of uncountable infinite state space for a univariate random variable θ . Then, the Metropolis-Hastings MCMC method is introduced for both the univariate and multivariate parameter spaces.

7.1 Markov Chain Monte Carlo (MCMC) - univariate case

To infer a parameter vector θ under a Bayesian setting, the key idea is to construct a Markov Chain whose stationary distribution is the posterior distribution $p(\theta|\mathbf{y})$. Subsequently, a realization of this Markov chain is generated. As the Markov chain converges to its stationary distribution (in this case, to the posterior distribution), the realization (after having removed the initial samples called **burn-in phase**) will be a posterior sample. This posterior sample can then be used to approximate all the quantities of interest of the posterior distribution.

For simplicity, we will start focusing on the case when the parameter space is univariate, that is the vector parameter θ reduces to a scalar θ . An MCMC simulation starts with the initialization of the unknown parameter, e.g. $\theta = \theta^{(1)}$. Then, the subsequent iterations are performed within a for-loop, defined as follows.

Each MCMC iteration $t \in \{1, \dots, 2T\}$ consists of two steps:

1. It is proposed to move from the current value $\theta^{(t)}$ of the unknown parameter to a new value θ^* . Often, this new candidate value is obtained by adding a small random number v (with expectation 0) to the current value $\theta^{(t)}$, for example

$$\theta^* = \theta^{(t)} + v, \quad v \sim U([- \epsilon, \epsilon]). \quad (116)$$

In general $\theta^* \sim Q(\theta^{(t)}, \theta^*)$, where Q is the proposal distribution.

2. The new value is either accepted or rejected according to some acceptance probability $A(\theta^{(t)}, \theta^*)$. If accepted, the next value is set to the candidate value, that is $\theta^{(t+1)} = \theta^*$. If rejected, the current value is left unchanged, that is $\theta^{(t+1)} = \theta^{(t)}$.

7.2 Metropolis-Hastings Algorithm - univariate case

In the Metropolis-Hastings MCMC method (univariate case), the previous algorithm becomes as follows:

1. **(Initialization):** set $\theta^{(1)} = \theta \in \Theta$, where $\theta^{(1)}$ can be any possible value from the parameter space Θ .
2. **(MCMC iterations):** for $t = 1, \dots, 2T$
 - (a) Propose to move from the current state $\theta^{(t)}$ to θ^* , where the candidate value θ^* is proposed with proposal probability $Q(\theta^{(t)}, \theta^*)$.
 - (b) Accept the new state θ^* with acceptance probability $A(\theta^{(t)}, \theta^*)$, that is
 - compute $A(\theta^{(t)}, \theta^*)$ consistent with Eq. (95);
 - draw a random number $u \sim U([0, 1])$
 - if $u \leq A(\theta^{(t)}, \theta^*)$, then set $\theta^{(t+1)} = \theta^*$ (accept the candidate)
 - if $u > A(\theta^{(t)}, \theta^*)$, then set $\theta^{(t+1)} = \theta^{(t)}$ (leave the state unchanged)
 - (c) Do the next iteration.

The proposal probabilities $Q(\theta^{(t)}, \theta^*)$

In the Metropolis-Hastings MCMC algorithm, the proposal probabilities $Q(\theta^{(t)}, \theta^*)$ in step 2(a) can be chosen arbitrarily, as long as two conditions are fulfilled:

1. $Q(x, y) > 0$ if and only if $Q(y, x) > 0$.
2. Every parameter $x \in \Theta$ must be reached from every parameter $y \in \Theta$, in a finite number of steps.

A standard approach was already mentioned in the previous section: given $\theta^{(t)}$, a random number v is sampled from a uniform distribution over the interval $[-\epsilon, \epsilon]$, that is $v \sim U([-\epsilon, \epsilon])$, and the new candidate value is defined as

$$\theta^* = \theta^{(t)} + v. \quad (117)$$

The acceptance probabilities $A(\theta^{(t)}, \theta^*)$

In the Metropolis-Hastings MCMC algorithm, the acceptance probabilities $A(\theta^{(t)}, \theta^*)$ in step 2(b) have to be computed consistent with Eq.(95), that is

$$A(\theta^{(t)}, \theta^*) = \min \left\{ 1, \frac{p(\theta^*|\mathbf{y})}{p(\theta^{(t)}|\mathbf{y})} \frac{Q(\theta^{(t)}, \theta^*)}{Q(\theta^*, \theta^{(t)})} \right\} \quad (118)$$

As we have seen in Proposition 7, this choice of $A(\theta^{(t)}, \theta^*)$ ensures that the target distribution, in this case the posterior distribution, fulfills the Equation of Detailed Balance (see Eq. (88)). This means that such a defined MCMC if simulated for “long enough” will “converge” to the posterior distribution $p(\theta|\mathbf{y})$ as its stationary distribution. Moreover, for the proposal mechanism described above, the Hastings Ratio $\frac{Q(\theta^*, \theta^{(t)})}{Q(\theta^{(t)}, \theta^*)}$ is always equal to 1. Therefore, the acceptance probability reduces to

$$A(\theta^{(t)}, \theta^*) = \min \left\{ 1, \frac{p(\theta^*|\mathbf{y})}{p(\theta^{(t)}|\mathbf{y})} \right\} \quad (119)$$

Furthermore, since

$$p(\theta|\mathbf{y}) \propto_\theta p(\mathbf{y}|\theta)p(\theta), \quad (120)$$

the acceptance probability further reduces to

$$A(\theta^{(t)}, \theta^*) = \min \left\{ 1, \frac{p(\mathbf{y}|\theta^*)p(\theta^*)}{p(\mathbf{y}|\theta^{(t)})p(\theta^{(t)})} \right\}. \quad (121)$$

7.3 Metropolis-Hastings Algorithm - multivariate case

In more realistic scenarios it is likely that the mathematical model under investigation has a multivariate parameter space Θ of dimension K . In this case, the univariate Metropolis-Hastings MCMC method can be extended as follows. Assume that the underlying model has an unknown K -dimensional vector parameter

$$\boldsymbol{\theta} = (\theta_1, \dots, \theta_K). \quad (122)$$

Then every MCMC iteration consists of K substeps. That is, in iteration t we move from the current state

$$(\theta_1^{(t)}, \dots, \theta_k^{(t)}, \theta_{k+1}^{(t)}, \dots, \theta_K^{(t)}) \quad (123)$$

to the new state

$$(\theta_1^{(t+1)}, \dots, \theta_k^{(t+1)}, \theta_{k+1}^{(t+1)}, \dots, \theta_K^{(t+1)}) \quad (124)$$

in K successive substeps:

- (step 1): Propose to go from $(\theta_1^{(t)}, \theta_2^{(t)}, \dots, \theta_k^{(t)}, \dots, \theta_K^{(t)})$ to $(\theta_1^*, \theta_2^{(t)}, \dots, \theta_k^{(t)}, \dots, \theta_K^{(t)})$
- (step 2): Propose to go from $(\theta_1^{(t+1)}, \theta_2^{(t)}, \theta_3^{(t)}, \dots, \theta_k^{(t)}, \dots, \theta_K^{(t)})$ to $(\theta_1^{(t+1)}, \theta_2^*, \theta_3^{(t)}, \dots, \theta_k^{(t)}, \dots, \theta_K^{(t)})$:
- (step k): Propose to go from $(\theta_1^{(t+1)}, \dots, \theta_{k-1}^{(t+1)}, \theta_k^{(t)}, \theta_{k+1}^{(t)}, \dots, \theta_K^{(t)})$ to $(\theta_1^{(t+1)}, \dots, \theta_{k-1}^{(t+1)}, \theta_k^*, \theta_{k+1}^{(t)}, \dots, \theta_K^{(t)})$
- (step K): Propose to go from $(\theta_1^{(t+1)}, \dots, \theta_{K-1}^{(t+1)}, \theta_K^{(t)})$ to $(\theta_1^{(t+1)}, \dots, \theta_{K-1}^{(t+1)}, \theta_K^*)$

where

$$\theta_k^{(t+1)} = \begin{cases} \theta_k^{(t)} & \text{if the } k\text{-th move is rejected} \\ \theta_k^* & \text{if the } k\text{-th move is accepted} \end{cases} \quad (125)$$

That is, the k -th step proposes to move from $\boldsymbol{\theta}_A$ to $\boldsymbol{\theta}_B$, where

$$\begin{aligned} \boldsymbol{\theta}_A &= (\theta_1^{(t+1)}, \dots, \theta_{k-1}^{(t+1)}, \theta_k^{(t)}, \theta_{k+1}^{(t)}, \dots, \theta_K^{(t)}) \\ \boldsymbol{\theta}_B &= (\theta_1^{(t+1)}, \dots, \theta_{k-1}^{(t+1)}, \theta_k^*, \theta_{k+1}^{(t)}, \dots, \theta_K^{(t)}), \end{aligned} \quad (126)$$

whose acceptance probability is

$$A(\boldsymbol{\theta}_A, \boldsymbol{\theta}_B) = \min \left\{ 1, \frac{p(\mathbf{y}|\boldsymbol{\theta}_B)p(\boldsymbol{\theta}_B)}{p(\mathbf{y}|\boldsymbol{\theta}_A)p(\boldsymbol{\theta}_A)} \right\}. \quad (127)$$

After $2T$ steps, the algorithm defined above will generate a posterior sample of the form

$$\begin{aligned} \boldsymbol{\theta}^{(1)} &= (\theta_1^{(1)}, \dots, \theta_K^{(1)}) \\ \boldsymbol{\theta}^{(2)} &= (\theta_1^{(2)}, \dots, \theta_K^{(2)}) \\ &\vdots \\ \boldsymbol{\theta}^{(2T)} &= (\theta_1^{(2T)}, \dots, \theta_K^{(2T)}). \end{aligned} \quad (128)$$

7.4 Gibbs sampling

When the **full conditional distribution** (fcd) of the k -th parameter θ_k can be computed, that is it has an explicit analytical form, then the k -th substep of the Metropolis-Hastings algorithm can be replaced by a more efficient Gibbs sampling step. The density of the fcd of the k -th parameter is defined as

$$p(\theta_k | \theta_1, \dots, \theta_{k-1}, \theta_{k+1}, \dots, \theta_K, \mathbf{y}) \quad (129)$$

or short

$$p(\theta_k | \boldsymbol{\theta}_{-k}, \mathbf{y}), \quad (130)$$

where $\boldsymbol{\theta}_{-k}$ denotes all parameters except for the k -th component. In the k -th substep of the Gibbs MCMC algorithm, instead of the Metropolis-Hastings move

$$\theta_k^* = \theta_k^{(t)} + v, \quad v \sim U([- \epsilon, \epsilon]), \quad (131)$$

we re-sample θ_k^* from its full conditional distribution. In this case, the acceptance probability $A(\boldsymbol{\theta}_A, \boldsymbol{\theta}_B)$ of the k -th submove is equal to 1. In fact, recall that the k -th substep proposes to move from $\boldsymbol{\theta}_A$ to $\boldsymbol{\theta}_B$, where

$$\begin{aligned} \boldsymbol{\theta}_A &= (\theta_1^{(t+1)}, \dots, \theta_{k-1}^{(t+1)}, \theta_k^{(t)}, \theta_{k+1}^{(t)}, \dots, \theta_K^{(t)}) \\ \boldsymbol{\theta}_B &= (\theta_1^{(t+1)}, \dots, \theta_{k-1}^{(t+1)}, \theta_k^*, \theta_{k+1}^{(t)}, \dots, \theta_K^{(t)}). \end{aligned} \quad (132)$$

Then, if we define

$$\boldsymbol{\theta}_{-k}^{(t)} := (\theta_1^{(t+1)}, \dots, \theta_{k-1}^{(t+1)}, \theta_{k+1}^{(t)}, \dots, \theta_K^{(t)}) \quad (133)$$

as the vector of the current states except the k -th component, the acceptance probability for the k -th step is

$$A(\boldsymbol{\theta}_A, \boldsymbol{\theta}_B) = \min \left\{ 1, \frac{p(\boldsymbol{\theta}_B | \mathbf{y}) Q(\boldsymbol{\theta}_B, \boldsymbol{\theta}_A)}{p(\boldsymbol{\theta}_A | \mathbf{y}) Q(\boldsymbol{\theta}_A, \boldsymbol{\theta}_B)} \right\}, \quad (134)$$

where the Hastings ratio reduces to

$$\begin{aligned} \frac{Q(\boldsymbol{\theta}_B, \boldsymbol{\theta}_A)}{Q(\boldsymbol{\theta}_A, \boldsymbol{\theta}_B)} &= \frac{p(\theta_k^{(t)} | \boldsymbol{\theta}_{-k}^{(t)}, \mathbf{y})}{p(\theta_k^* | \boldsymbol{\theta}_{-k}^{(t)}, \mathbf{y})} \\ &= \frac{\frac{p(\theta_k^{(t)}, \boldsymbol{\theta}_{-k}^{(t)} | \mathbf{y})}{p(\boldsymbol{\theta}_{-k}^{(t)} | \mathbf{y})}}{\frac{p(\theta_k^*, \boldsymbol{\theta}_{-k}^{(t)} | \mathbf{y})}{p(\boldsymbol{\theta}_{-k}^{(t)} | \mathbf{y})}} \\ &= \frac{p(\theta_k^{(t)}, \boldsymbol{\theta}_{-k}^{(t)} | \mathbf{y})}{p(\theta_k^*, \boldsymbol{\theta}_{-k}^{(t)} | \mathbf{y})} \\ &= \frac{p(\boldsymbol{\theta}_A, \mathbf{y})}{p(\boldsymbol{\theta}_B, \mathbf{y})} \\ &= \frac{p(\boldsymbol{\theta}_A | \mathbf{y}) p(\mathbf{y})}{p(\boldsymbol{\theta}_B | \mathbf{y}) p(\mathbf{y})} \\ &= \frac{p(\boldsymbol{\theta}_A | \mathbf{y})}{p(\boldsymbol{\theta}_B | \mathbf{y})}. \end{aligned} \quad (135)$$

Therefore,

$$\begin{aligned} A(\boldsymbol{\theta}_A, \boldsymbol{\theta}_B) &= \min \left\{ 1, \frac{p(\boldsymbol{\theta}_B | \mathbf{y}) Q(\boldsymbol{\theta}_B, \boldsymbol{\theta}_A)}{p(\boldsymbol{\theta}_A | \mathbf{y}) Q(\boldsymbol{\theta}_A, \boldsymbol{\theta}_B)} \right\} \\ &= \min \left\{ 1, \frac{p(\boldsymbol{\theta}_B | \mathbf{y}) p(\boldsymbol{\theta}_A | \mathbf{y})}{p(\boldsymbol{\theta}_A | \mathbf{y}) p(\boldsymbol{\theta}_B | \mathbf{y})} \right\} \\ &= 1. \end{aligned} \quad (136)$$

Remark 2. Instead of $\frac{Q(\boldsymbol{\theta}_B, \boldsymbol{\theta}_A)}{Q(\boldsymbol{\theta}_A, \boldsymbol{\theta}_B)}$ we could simply write $\frac{Q(\theta_k^*, \theta_k^{(t)})}{Q(\theta_k^{(t)}, \theta_k^*)}$, as only the k -th parameter changes. Furthermore, since the new state of the k -th parameter is sampled from its full conditional distribution, it does not even depend on its current state. This means that the Hastings ratio can be further written as $\frac{Q(\theta_k^{(t)})}{Q(\theta_k^*)}$.

7.5 Example

7.6 Bayesian Uncertainty Quantification

Assume that we want to estimate the parameters θ of a given model $\mathcal{M}_\theta$, based on observed experimental data. As mentioned earlier, in a Bayesian framework first a prior distribution $p(\theta)$ is imposed on the vector parameter θ . Then, given some data $\mathbf{y}$ and a likelihood function $p(\mathbf{y}|\theta)$ (typically based on a “distance” between observed data and the corresponding model predictions), we can update our knowledge about θ after having observed data by computing the posterior distribution

$$p(\theta|\mathbf{y}) \propto_\theta p(\mathbf{y}|\theta)p(\theta). \quad (137)$$

If we knew the exact analytical form of the posterior, we could have draw conclusion about some quantities of interest regarding θ , such as the mean, its variance, and confidence intervals/regions. However, often the posterior $p(\theta|\mathbf{y})$ is unknown but can be approximated by MCMC samples. In fact, once such samples are generated, for example via the Metropolis-Hastings MCMC method, the properties of interest of the posterior distribution can be empirically approximated using sample statistics (e.g. the sample mean, sample median, sample variance, confidence intervals, etc.).

It is also worth to note that once a MCMC sample $\{\theta^{(t)}\}_t$ is generated from the multivariate posterior distribution $p(\theta|\mathbf{y})$, each marginal MCMC sample $\{\theta_j^{(t)}\}_t$ can be used to approximate properties of the posterior distribution $p(\theta_j|\mathbf{y})$ of the j -th component θ_j of θ . In the following, we present two techniques to build confidence intervals for each component θ_j of θ , based on their posterior distributions $p(\theta_j|\mathbf{y})$.

Quantile-based confidence intervals

Perhaps the easiest way to obtain a confidence interval of a scalar parameter θ is to use the quantiles of its posterior distribution. At a level α , to make a $100 \times (1 - \alpha)\%$ quantile-based confidence interval for θ , it is required to find the $\alpha/2$ and $1 - \alpha/2$ posterior quantiles of θ , that is the numbers $\theta_{\alpha/2} < \theta_{1-\alpha/2}$ satisfying

$$\begin{aligned} P(\theta < \theta_{\alpha/2}|\mathbf{y}) &= \alpha/2 \\ P(\theta > \theta_{1-\alpha/2}|\mathbf{y}) &= \alpha/2. \end{aligned} \quad (138)$$

This means that $\theta_{\alpha/2}$ and $\theta_{1-\alpha/2}$ also satisfy

$$\begin{aligned} P(\theta \in [\theta_{\alpha/2}, \theta_{1-\alpha/2}]|\mathbf{y}) &= 1 - P(\theta \notin [\theta_{\alpha/2}, \theta_{1-\alpha/2}]|\mathbf{y}) \\ &= 1 - [P(\theta < \theta_{\alpha/2}|\mathbf{y}) + P(\theta > \theta_{1-\alpha/2}|\mathbf{y})] \\ &= 1 - [\alpha/2 + \alpha/2] = 1 - \alpha. \end{aligned} \quad (139)$$

Therefore, $\theta_{\alpha/2}$ and $\theta_{1-\alpha/2}$ define the lower and upper bounds of a $100 \times (1 - \alpha)\%$ quantile-based confidence interval for θ . One limitation of this method is that there might be θ values outside the quantile-based interval that have a higher posterior probability density than some points inside the interval, as shown in the example of Fig. 1.

Highest posterior density region

We have seen earlier that a quantile-based confidence interval can contain θ values with lower posterior probability density than some value outside the confidence interval. This limitation can be overcome by a more restrictive type of confidence interval.

Definition 9. Given a level α , a $100 \times (1 - \alpha)\%$ **highest posterior density (HPD) region** for a parameter $\theta \in \Theta$, consists of a subset $S_{\mathbf{y}} \subset \Theta$ of the parameter space Θ , such that

1. $P(\theta \in S_{\mathbf{y}}|\mathbf{y}) = 1 - \alpha$
2. if $\theta_a \in S_{\mathbf{y}}$ and $\theta_b \notin S_{\mathbf{y}}$, then $P(\theta_a|\mathbf{y}) > P(\theta_b|\mathbf{y})$.

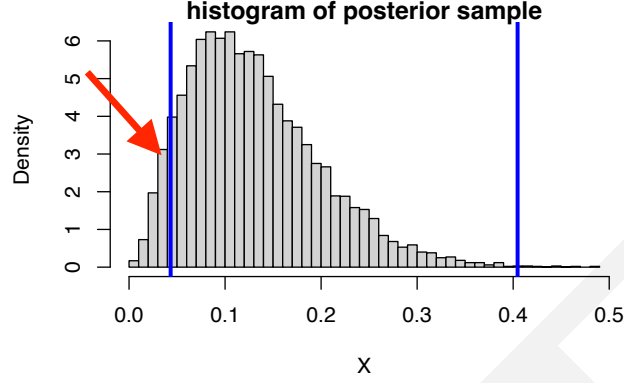


Figure 1: An illustrative example of histogram of a MCMC posterior sample of θ . The vertical blue lines indicate the lower and upper bounds of a 5% quantile-based confidence interval. The red arrow points to a θ value outside the confidence interval that has a higher posterior probability density than some points inside the interval.

Therefore, all points in an HPD region have a higher posterior density than points outside the region. However, an HPD region might not be an interval if the posterior density is multimodal (that is, having multiple peaks). This is why we refer to it as region instead of as an interval.

An HPD region can be constructed by

- Gradually move a horizontal line down across the density, including in the HPD region all the θ values having a density above the horizontal line.
- Stop moving the line down when the posterior probability of the θ values in the HPD region reaches $(1 - \alpha)$.

An HPD region is computed for the posterior sample of previous example, and the result is displayed in Fig. 2, where a comparison is made with the quantile-based confidence interval previously computed.

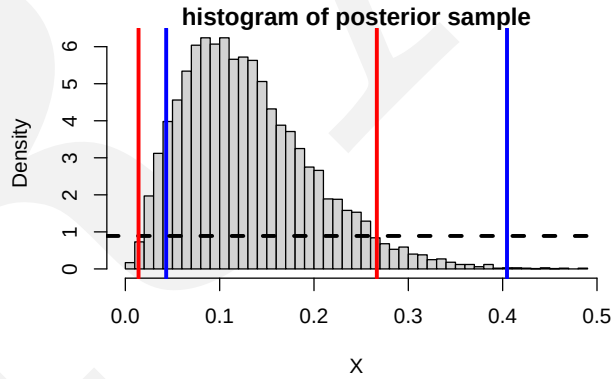


Figure 2: An illustrative example of histogram of a MCMC posterior sample of θ . The vertical blue lines indicate the lower and upper bounds of a 5% quantile-based confidence interval. The vertical red lines indicate the lower and upper bounds of a 5% HPD region, identified by the horizontal dashed black line.

7.7 Posterior Predictive Inference

After obtaining an MCMC sample from the posterior distribution

$$p(\boldsymbol{\theta} \mid \mathbf{y}), \quad (140)$$

where $\boldsymbol{\theta}$ denotes the model parameters and $\mathbf{y}$ the observed data, we can make predictions about new or unobserved data points $\tilde{y}$. The posterior predictive distribution is defined as

$$p(\tilde{y} | \mathbf{y}) = \int p(\tilde{y} | \boldsymbol{\theta}) p(\boldsymbol{\theta} | \mathbf{y}) d\boldsymbol{\theta}. \quad (141)$$

This integral represents an average of the likelihood of $\tilde{y}$ over the posterior uncertainty in $\boldsymbol{\theta}$. Computing it analytically is often intractable, but MCMC samples $\{\boldsymbol{\theta}^{(s)}\}_{s=1}^S$ from the posterior provide a natural Monte Carlo approximation. Specifically, by the law of large numbers, if the draws $\boldsymbol{\theta}^{(s)}$ are (approximately) independent samples from $p(\boldsymbol{\theta} | \mathbf{y})$, we have

$$\int p(\tilde{y} | \boldsymbol{\theta}) p(\boldsymbol{\theta} | \mathbf{y}) d\boldsymbol{\theta} \approx \frac{1}{S} \sum_{s=1}^S p(\tilde{y} | \boldsymbol{\theta}^{(s)}). \quad (142)$$

The equation above approximates the density at a fixed $\tilde{y}$. However, in practice, we often want actual predictive samples rather than the density itself. In other words, the two-step procedure defined by

1. Each $\boldsymbol{\theta}^{(s)}$ is drawn from $p(\boldsymbol{\theta} | \mathbf{y})$,
2. Conditional on $\boldsymbol{\theta}^{(s)}$, the new data point $\tilde{y}$ has the distribution $p(\tilde{y} | \boldsymbol{\theta})$,

is equivalent to drawing $\tilde{y}$ from the mixture distribution defined by the integral of Eq. (141), or its Monte Carlo approximation given by Eq. (142).

Intuitively, each posterior draw $\boldsymbol{\theta}^{(s)}$ represents a plausible parameter configuration given the observed data. Averaging over these draws approximates integrating over all plausible parameter values weighted by their posterior probability. In practice, to generate a posterior predictive sample, for each MCMC draw $\boldsymbol{\theta}^{(s)}$, we simulate

$$\tilde{y}^{(s)} \sim p(\tilde{y} | \boldsymbol{\theta}^{(s)}). \quad (143)$$

The collection of simulated values $\{\tilde{y}^{(s)}\}_{s=1}^S$ provides an empirical approximation of the posterior predictive distribution. Prediction intervals for $\tilde{y}$ can then be obtained by taking quantiles of this sample. For example, a 95% predictive interval is given by the 2.5th and 97.5th percentiles of $\{\tilde{y}^{(s)}\}$. This approach naturally accounts for both parameter uncertainty (through the posterior $p(\boldsymbol{\theta} | \mathbf{y})$) and inherent randomness in the data (through $p(\tilde{y} | \boldsymbol{\theta})$), giving a coherent Bayesian measure of uncertainty for future observations.

8 State-space modelling of dynamical systems

Since the aim of uncertainty quantification of dynamical systems is to estimate the parameters, the model output, and their uncertainty, of a continuous-time mathematical model from experimental data that have been recorded at discrete time points, we consider a continuous-discrete state-space formulation of the dynamical systems. Such formulation naturally links the discrete-time measurements to the continuous-time mathematical formulation of the process under investigation [9]. In particular, we consider a state-space formulation including a dynamic model defined as a system of ordinary differential equations (ODEs)

$$\frac{d\mathbf{x}}{dt} = f(\mathbf{x}(t), \mathbf{u}(t), \boldsymbol{\theta}), \quad (144)$$

called the **dynamic model**, coupled with a **measurement model**

$$y_t = h(\mathbf{x}_t, \boldsymbol{\alpha}) + \varepsilon, \quad \varepsilon \sim \mathcal{N}(0, \sigma^2), \quad (145)$$

where $\mathbf{x}(t) \in \mathbb{R}^{n_x}$ is the state vector of the system, $\mathbf{u}(t) \in \mathbb{R}^{n_u}$ is the model input, for example an external stimulus, $\boldsymbol{\theta} \in \Theta \subseteq \mathbb{R}^p$ is the vector parameter, $h(\cdot)$ is the **measurement function**, parametrized by $\boldsymbol{\alpha} \in A \subseteq \mathbb{R}^q$, linking the state vector $\mathbf{x}(t)$ to the experimental measurement $y_t \in \mathbb{R}^{n_y}$, where typically $n_y \leq n_x$. Finally ε is the measurement noise. It is worth to note that typically the residuals $y_t - h(\mathbf{x}_t) = \varepsilon$ are assumed homoscedastic, independent and Gaussian distributed. However, this assumption might be too simplistic and can be relaxed if required. It is also worth to note that usually the measurement parameters $\boldsymbol{\alpha}$ are given, whereas the dynamic parameters $\boldsymbol{\theta}$ are unknown and have to be estimated from experimental data.

8.1 Structural and Practical Identifiability

When modelling a dynamic system, its behaviour is often described by a set of differential equations that depend on unknown parameters, such as reaction rate constants, growth rates, or diffusion coefficients. To make reliable predictions or design effective control strategies, it is necessary to estimate these parameters from experimental or observational data. This raises a fundamental question: *can the parameters actually be determined from the data we can observe?* This question is addressed by the concepts of structural and practical identifiability.

Structural identifiability concerns the model structure itself. It asks whether, in theory, perfect and noise-free measurements of the system outputs would uniquely determine the parameter values. If two or more distinct parameter sets can produce exactly the same output trajectories for all admissible inputs, those parameters are said to be structurally unidentifiable. In such cases, no amount of improved data quality can resolve the ambiguity, and the model must be reformulated (for example, by reparameterization or by measuring additional outputs).

Practical identifiability, in contrast, addresses the limitations of real-world data. Even if a model is structurally identifiable, the available experimental data are typically finite, noisy, and may not sufficiently excite all system dynamics. Practical identifiability therefore assesses whether the data are informative enough to estimate the parameters with acceptable accuracy and confidence. Parameters that are theoretically identifiable but yield very wide confidence intervals or highly correlated estimates are considered practically unidentifiable.

More formally, consider a parametric dynamic system defined by the state-space formulation of Eqs. (144–145), where we assume that the input function $\mathbf{u}(t)$ and the measurement parameters $\boldsymbol{\alpha}$ are known and given.

Definition 10. *The model is said to be **structurally identifiable** if, under the assumption of perfect (noise-free, continuous, and complete) data, the mapping from the dynamic parameter vector $\boldsymbol{\theta}$ to the model output $y_t = h(\mathbf{x}(t, \boldsymbol{\theta}))$ is injective. More formally, the model is **globally structurally identifiable** at $\boldsymbol{\theta}^* \in \Theta$ if,*

$$\forall \boldsymbol{\theta} \in \Theta, h(\mathbf{x}(t, \boldsymbol{\theta})) = h(\mathbf{x}(t, \boldsymbol{\theta}^*)) \quad \forall t \quad \Rightarrow \quad \boldsymbol{\theta} = \boldsymbol{\theta}^*, \quad (146)$$

where Θ is the vector parameter space. Furthermore, the model is **locally structurally identifiable** if the implication defined above holds only in a neighborhood $\mathcal{I}_{\boldsymbol{\theta}^*}$ of $\boldsymbol{\theta}^*$. As a consequence, if there exist distinct parameter vectors $\boldsymbol{\theta}_1 \neq \boldsymbol{\theta}_2$ such that $h(\mathbf{x}(t, \boldsymbol{\theta}_1)) = h(\mathbf{x}(t, \boldsymbol{\theta}_2))$ for all admissible inputs $\mathbf{u}(t)$, then the model is referred to as **structurally unidentifiable**.

It is worth to note that structural identifiability depends solely on the model structure and not on the quality or quantity of data, nor on the experimental design. This is why structural identifiability it is also called ***a priori identifiability***. If a model results in being structurally unidentifiable, it is good practice to modify its formulation so as to obtain a new identifiable mathematical model which is priori identifiable.

Definition 11. *Given a structurally identifiable state-space model, the dynamic parameters θ are said to be **practically identifiable** if, for a given finite set of experimental data $\mathbf{y} = (y_1, y_2, \dots)'$, the parameter estimates can be determined with finite confidence and acceptable precision. More formally, the dynamic parameters θ are practically identifiable if the (measurement) model log-likelihood $\ell(\mathbf{y}|\theta)$ exhibits a unique and well-defined maximum, i.e., it is sharply peaked.*

It is worth to note that while structural identifiability is a theoretical property of a model that determines whether its parameters can, in principle, be uniquely recovered from perfect, noise-free, and continuous observations, practical identifiability, on the other hand, is a data-dependent property that assesses whether parameters can be reliably estimated in practice, given real, noisy, and finite experimental data. It depends strongly on data quality, experimental design, noise level, and parameter sensitivity.

9 Numerical examples

9.1 A logistic model with unknown initial condition and growth rate

To evaluate the performance of several uncertainty quantification methods, we use a numerical example based on a linear regression model. More specifically, consider a state space model whose dynamics is described by a logistic model

$$\frac{dx}{dt} = \lambda x(1 - x), \quad x \in (0, 1), \quad (147)$$

with initial condition $x_0 = x(0)$ and growth rate λ , whose solution is given by

$$x(t) = \frac{x_0 e^{\lambda t}}{(1 - x_0) + x_0 e^{\lambda t}}. \quad (148)$$

Taking a logit transformation of both sides of the above, gives

$$\begin{aligned} \log \frac{x}{1 - x} &= \log \frac{x_0}{1 - x_0} + \lambda t, \\ &= \beta_0 + \beta_1 t, \end{aligned} \quad (149)$$

that is a linear model in β_0 and β_1 , where

$$\beta_0 := \log \frac{x_0}{1 - x_0}, \quad \beta_1 := \lambda. \quad (150)$$

This means that, if we have measurements x_i at time t_i , we can transform them as

$$y_i = \log \frac{x_i}{1 - x_i}, \quad (151)$$

and perform a linear regression of y_i versus t_i ,

$$y_i = \beta_0 + \beta_1 t_i + \varepsilon_i, \quad \varepsilon_i \sim \mathcal{N}(0, \sigma^2). \quad (152)$$

to infer $\hat{\beta}_0$ and $\hat{\beta}_1$ via ordinary least squares, where we assumed that residuals are iid and Gaussian distributed. Finally, we can obtain an estimate of the initial condition x_0 and the growth rate λ as

$$\hat{x}_0 = \frac{e^{\hat{\beta}_0}}{1 + e^{\hat{\beta}_0}}, \quad \hat{\lambda} = \hat{\beta}_1. \quad (153)$$

It is worth to note that Eqs. (147–152) form a state-space model, consistent with the general formulation of Eqs. (144–145). In the following, we use the following matrix formulation

$$Y = X + \varepsilon, \quad (154)$$

for the regression problem of Eq. (152), where

$$Y = (y_1, \dots, y_n)' \quad (155)$$

is the vector of the measurements collected at times $t_1, \dots, t_n$,

$$X = \begin{bmatrix} 1 & t_1 \\ \vdots & \vdots \\ 1 & t_n \end{bmatrix} \quad (156)$$

is the design matrix, and

$$\varepsilon = (\varepsilon_1, \dots, \varepsilon_n)' \quad (157)$$

is the vector of the residuals.

	x_0	λ	σ^2
true	0.00050000	0.30000000	0.10000000
OLS	0.00049831	0.30049719	0.00836374
MCMC mean	0.00049798	0.30051692	0.00857647

Table 1: True values and OLS/Bayesian estimates (rows) for each parameter (columns) for the true data generating process specified by Eqs. (147–152).

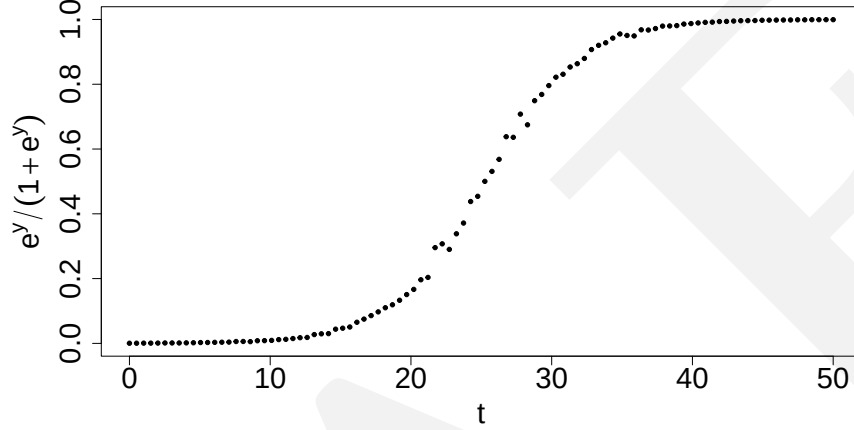


Figure 3: Scatterplot of the simulated response variable (y -axis), on a logistic scale, against time (x -axis).

Using the model specified by Eqs. (147–152), we generated $n = 100$ synthetic data points (t_i, y_i) , with the true parameters set according to Table 9.1. The time points t_i were chosen as a sequence of 100 values evenly spaced over the interval $(0, 50)$. The generated synthetic data is displayed in Fig. 3. Subsequently, we compute the OLS estimator of $\boldsymbol{\beta} = (\beta_0, \beta_1)'$ as

$$\hat{\boldsymbol{\beta}} = (X'X)^{-1}X'Y, \quad (158)$$

and an unbiased sample estimate for the noise variance σ^2 as

$$\hat{\sigma}^2 = \frac{1}{n-p}(Y - X\hat{\boldsymbol{\beta}})'(Y - X\hat{\boldsymbol{\beta}}), \quad (159)$$

where $n-p$ is the degrees of freedom, and $p = 2$ the dimension of the unknown regression parameters $\boldsymbol{\beta}$. The estimated values are reported in Table 9.1, where $\hat{x}_0$ is obtained by using the transformation of Eq. (153). The Fisher Information matrix parabola and the profile likelihood were obtained consistent with Sections 2,3, and the results are displayed in Figs. 4 for a direct comparison. We also performed a Bayesian inference analysis on the same simulated data. Under this inference setting, we assumed that the unknown parameters $\boldsymbol{\beta}$ and σ^2 have the following conjugate prior distributions

$$\boldsymbol{\beta}|\sigma^2 \sim \mathcal{N}_2(\mathbf{0}, \mathbf{V}_0), \quad \sigma^2 \sim \text{Inverse-Gamma}(a_0, b_0), \quad (160)$$

so that the full conditional posteriors of $\boldsymbol{\beta}$ and σ^2 have explicit analytical form given by

$$\begin{aligned} \boldsymbol{\beta}|\sigma^2, Y &\sim \mathcal{N}_2(\boldsymbol{\mu}_p, \mathbf{V}_p), \\ \boldsymbol{\mu}_p &= \mathbf{V}_p X'Y/\sigma^2, \quad \mathbf{V}_p = (X'X/\sigma^2 + \mathbf{V}_0^{-1})^{-1}, \end{aligned} \quad (161)$$

and

$$\begin{aligned} \sigma^2|\boldsymbol{\beta}, Y &\sim \text{Inverse-Gamma}(a_p, b_p), \\ a_p &= a_0 + n/2, \quad b_p = b_0 + \frac{1}{2}(Y - X\boldsymbol{\beta})'(Y - X\boldsymbol{\beta}), \end{aligned} \quad (162)$$

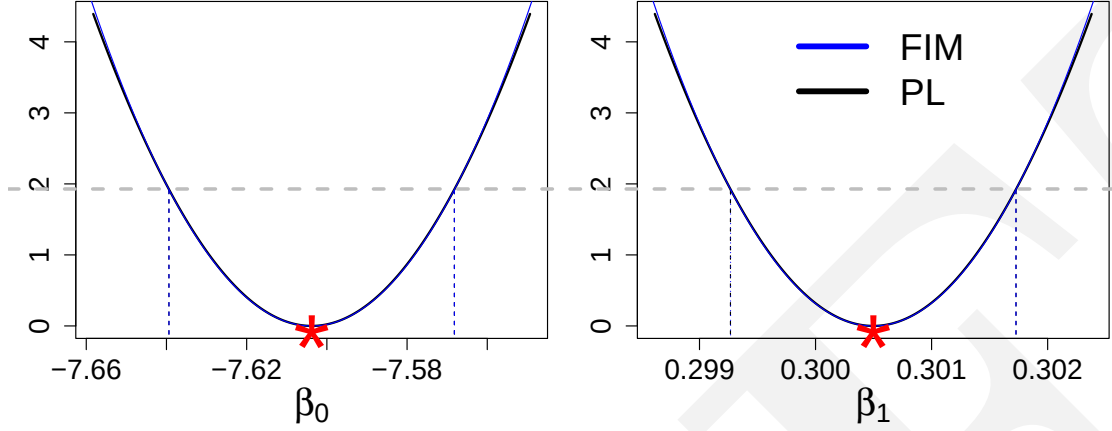


Figure 4: Fisher Information parabolas (blue) and Profile Likelihood (black), both on a negative loglikelihood scale, for the intercept β_0 (left) and slope β_1 (right). The horizontal grey line is the threshold $\ell(\hat{\beta}) - \frac{1}{2}\chi^2_{1-\alpha}$, $\alpha = 0.05$, used to obtain the $(1 - \alpha)$ confidence intervals (horizontal dashed lines) for both the FIM (blue) and PL (black) approaches. The maximum likelihood estimate $\hat{\beta}$ is indicated by a red star symbol.

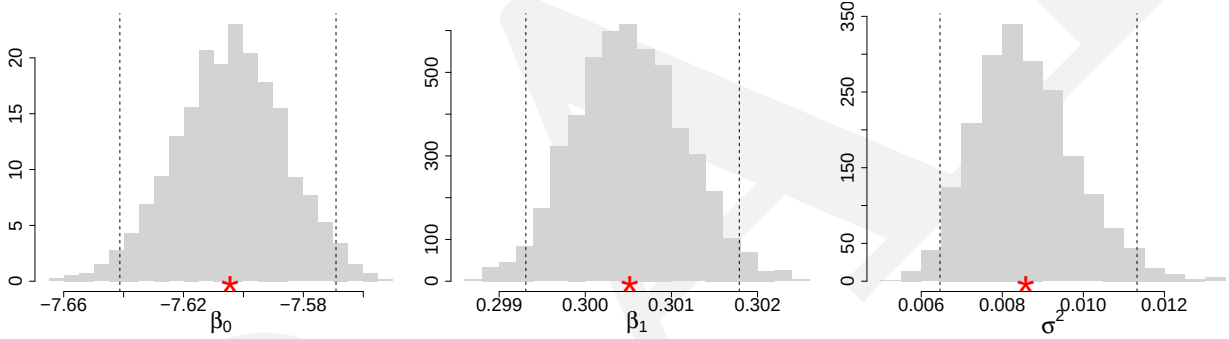


Figure 5: Histogram of the posterior MCMC samples for the intercept β_0 (left), the slope β_1 (middle), and the measurement noise variance σ^2 (right), obtained with the Gibbs sampling algorithm of Section ??, and prior-posterior given by Eqs. (160–162), with a burning-in period of 50%. The posterior means of β and σ^2 are indicated by red star symbols, whereas the corresponding 95% credible intervals are indicated by horizontal dashed lines.

respectively [10]. Then we generate $N = 5000$ MCMC samples from the posterior using the Gibbs sampling algorithm of Section 7.4, and prior-posterior given by Eqs. (160–162), with a burning-in period of 50%. The MCMC posterior samples are displayed in the histograms of Fig.5, including 95% credible intervals computed consistent with Section 7.6. Finally, we compute the prediction intervals for the response variable Y using the Fisher Information, the Profile Likelihood, the Bayesian posterior, and the jackknife+ Conformal prediction methods. The prediction intervals, together with the simulated synthetic data and the mean predictions obtained with the MLE and the Bayesian posterior mean, are displayed in Fig. 6. For a computational comparison, the runtime of each method is reported in Table 9.1.

As a result, Ordinary least squares (OLS) estimation closely recovered the true parameters: $\hat{\beta}_0$ and $\hat{\beta}_1$ were nearly unbiased and the estimated residual variance $\hat{\sigma}^2$ was close to the true $\sigma^2 = 0.01$, as reported in Table 9.1. This is expected because the assumed model matches the data-generating mechanism and the noise level is low. For linear regression with Gaussian errors, the log-likelihood is exactly quadratic in the parameters, and therefore the Fisher-based confidence intervals (CIs) serve as a natural benchmark in this context. The profile likelihood curves were nearly perfectly parabolic, and the resulting 95% profile likelihood intervals were virtually identical to the Fisher information intervals, as suggested by Figure 4. This agreement confirms that the model behaves in a regular fashion and that the Gaussian approximation

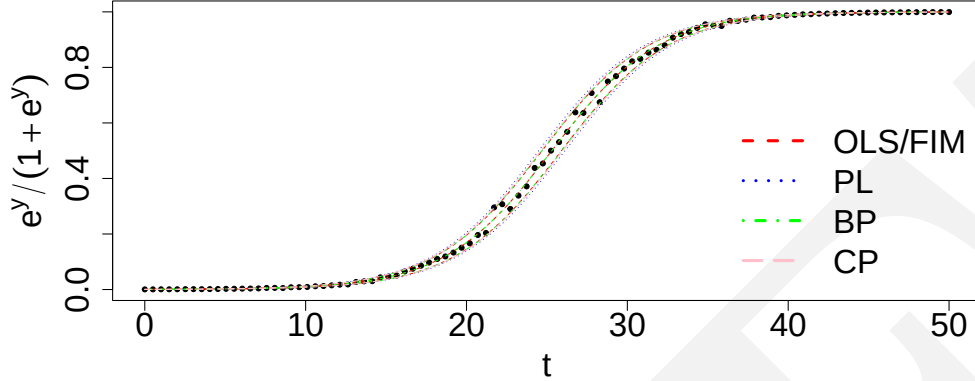


Figure 6: Scatterplot of synthetic data (dots) on a logistic scale (y -axis), over time (x -axis), the mean predictions obtained with the OLS estimates (red) and the MCMC Bayesian posterior sample means (green). The plot also includes the prediction intervals obtained with the Fisher information matrix (red), the Profile Likelihood (blue), the Bayesian posterior (green), and the Jackknife+ conformal prediction method (pink).

method	FIM	PL	Bayesian	Jackknife+ CP
time (s)	0.004	0.212	1.543	0.463

Table 2: Runtime (in seconds) for each UQ method (columns).

is fully adequate under these conditions.

The Bayesian MCMC posterior samples of β were approximately Gaussian, centered near the OLS estimates with variances matching the Fisher information estimates, consistent with the estimates reported in Table 9.1. Posterior samples of σ^2 exhibited the expected right-skewed inverse-gamma shape. The 95% Bayesian credible intervals for β_0 and β_1 were nearly indistinguishable from both the Fisher and profile likelihood intervals (see Figure 4), again reflecting the conjugate linear-Gaussian structure of the model. The Bayesian posterior predictive intervals were slightly wider than those derived from Fisher information or profile likelihood (see Fig. 6). This widening is attributable to integration over posterior uncertainty in both β and σ^2 . Nevertheless, the Bayesian predictive intervals were still very similar in shape and width to the parametric frequentist intervals. The 95% prediction intervals obtained via the Jackknife+ method were consistently wider than all parametric intervals, as suggested by Fig. 6. This result is consistent with the theoretical guarantee that conformal methods achieve nominal marginal coverage without assuming linearity, Gaussianity, or correct model specification. Despite being slightly wider, the Jackknife+ intervals closely tracked the sigmoidal trend of the mean function. Finally, from a computational point of view the Fisher information intervals were the fastest to run due to their reliance on closed-form expressions, consistent with the runtime reported in Table 9.1. Profile likelihood intervals required moderate computation because of repeated optimization over the nuisance parameters. Bayesian inference was substantially more expensive, primarily due to the cost of running thousands of Gibbs iterations. The Jackknife+ method fell between Fisher and Bayesian approaches in computational burden.

Overall, the results demonstrate that (1) all parametric methods produce nearly identical uncertainty estimates for this well-specified linear-Gaussian model; (2) Bayesian and frequentist approaches exhibit excellent agreement; and (3) conformal prediction provides valid intervals at the cost of increased width, while requiring no distributional assumptions. However, It is important to note that this experiment represents an intentionally simple, well-specified, and noise-controlled setting designed only to illustrate how these UQ methods can be implemented numerically. In more complex or misspecified models—such as nonlinear dynamics, heteroscedastic noise structures, latent variable models, or non-Gaussian error distributions—different UQ approaches may diverge more noticeably in both interval width and coverage properties. Thus, the apparent agreement observed here should not be interpreted as generally guaranteed, but rather as a feature of this toy linear example.

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